TCS PLACEMENT

MASTER PREPARATION GUIDE

NQT • Digital • Prime

■ 60-DAY COMPLETE STUDY PLAN ■

500+	6	60	4
Questions Covered	Core Subjects	Day Study Plan	Tier Priority System

DSA	SQL	DBMS	Operating Systems	Computer Networks	OOPs

Every question, concept and topic reported from TCS NQT / Digital / Prime interviews (2021-2026) — verified from GeeksforGeeks, PrepInsta, FacePrep, Reddit, GitHub, LinkedIn and YouTube interview experiences.

HOW TO USE THIS GUIDE

1 **Follow in Order**
Work through the 60-day plan sequentially. Each day has a fixed topic allocation so nothing is missed.

2 **Use Checkboxes**
Every question has a [ ] checkbox. Tick it off once you have coded/answered it confidently. Revisit empty boxes weekly.

3 **Respect Tiers**
Tier 1 = Absolute must. Tier 2 = Very important. Tier 3 = Good to know. Tier 4 = Rarely asked. Focus Tier 1+2 first.

4 **Frequency Matters**
Very High / High frequency questions appear in nearly every TCS screening. Master them completely before moving on.

5 **Tables Give Context**
Each question table shows the Round (NQT/Digital/Prime) so you tailor depth based on the level you are targeting.

6 **Top-N Lists**
Use the Top 25 / 50 / 100 quick-reference lists before mock tests for rapid revision.

60-Day Study Schedule

Period	Topic Focus	Strategy
Days 1-5	DSA - Arrays, Strings, Hashing	Tier 1 problems only
Days 6-10	DSA - Linked List, Stack, Queue	Tier 1+2 problems
Days 11-16	DSA - Trees (BST, Binary, Tries)	Tier 1+2 problems
Days 17-22	DSA - Graphs (BFS/DFS/Shortest Path)	Tier 1+2 problems
Days 23-27	DSA - DP (Classic patterns)	Tier 1 mandatory
Days 28-30	DSA - Sorting, Binary Search, Bit Manip	All Tiers
Days 31-35	SQL - All topics end-to-end	Write every query
Days 36-40	DBMS - Normalization, Keys, ACID, Locks	Tier 1+2
Days 41-45	OS - Scheduling, Memory, Deadlocks, Sync	Tier 1+2
Days 46-50	CN - OSI, TCP/IP, DNS, HTTP, Routing	Tier 1+2
Days 51-55	OOPs - All 4 pillars, SOLID, Patterns	Tier 1+2
Days 56-58	Full Revision - Top 100 All Subjects	Speed run
Days 59-60	Mock Tests + Weak Area Targeting	Simulate TCS exam

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SECTION 1 DSA

Data Structures & Algorithms — Every question reported from TCS NQT / Digital / Prime (2021-2026). Grouped by topic with frequency, difficulty and round information.

Arrays (48 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Find Second Largest Element**	Very High	Easy	NQT/Digital	YES	*Traverse array, track max & second max*
2	**Find Missing Number (1 to N)**	Very High	Easy	NQT	YES	*Sum formula or XOR trick*
3	**Leaders in an Array**	High	Easy	NQT/Digital	YES	*Element greater than all elements to its right*
4	**Kadane's Algorithm – Maximum Subarray Sum**	Very High	Medium	Digital/Prime	YES	*DP-based O(n) approach*
5	**Move All Zeroes to End**	High	Easy	NQT	YES	*Two-pointer in-place*
6	**Find Duplicates in Array**	High	Easy	NQT/Digital	YES	*Hashing or sorting*
7	**Rotate Array by K positions**	High	Easy	NQT	YES	*Reverse trick*
8	**Find Pair with Given Sum (Two Sum)**	Very High	Easy	NQT/Digital	YES	*HashMap O(n)*
9	**Find Triplet with Given Sum (3-Sum)**	High	Medium	Digital/Prime	YES	*Sort + two pointer*
10	**Merge Two Sorted Arrays**	High	Easy	NQT/Digital	YES	*Two-pointer merge*
11	**Find Intersection of Two Arrays**	Medium	Easy	NQT	YES	*Set / sorting*
12	**Find Union of Two Arrays**	Medium	Easy	NQT	YES	*Merge approach*
13	**Equilibrium Index of Array**	Medium	Easy	NQT	YES	*Prefix sum*
14	**Maximum Product Subarray**	High	Medium	Digital/Prime	YES	*Track max/min running product*
15	**Subarray with Given Sum**	High	Medium	NQT/Digital	YES	*Sliding window*
16	**Count Inversions in Array**	Medium	Medium	Digital/Prime	YES	*Modified merge sort*
17	**Best Time to Buy and Sell Stock**	Very High	Easy	Digital/Prime	YES	*Single pass min tracking*

#	Question	Freq	Diff	Round	Rep?	Description
18	**Best Time to Buy and Sell Stock II (multiple transactions)**	Medium	Medium	Prime	YES	*Greedy, add all rises*
19	**Rearrange Array in Alternating +/- Sign**	Medium	Medium	Digital	YES	*Partition then merge*
20	**Find All Subarrays with Zero Sum**	Medium	Medium	Digital	YES	*Prefix sum + HashMap*
21	**Longest Subarray with Equal 0s and 1s**	Medium	Medium	Digital	YES	*Convert 0->-1, prefix sum*
22	**Find Minimum in Rotated Sorted Array**	High	Medium	Digital/Prime	YES	*Binary search*
23	**Search in Rotated Sorted Array**	High	Medium	Digital/Prime	YES	*Modified binary search*
24	**Merge Intervals**	High	Medium	Digital/Prime	YES	*Sort + merge overlaps*
25	**Maximum Circular Subarray Sum**	Medium	Hard	Prime	YES	*Kadane + total sum trick*
26	**Sliding Window Maximum**	Medium	Hard	Prime	no	*Deque-based O(n)*
27	**Trapping Rain Water**	High	Hard	Digital/Prime	YES	*Two-pointer or stack*
28	**Count pairs with given difference K**	Medium	Easy	NQT	YES	*HashSet lookup*
29	**Array Rotation – Juggling Algorithm**	Medium	Medium	Digital	no	*GCD-based rotation*
30	**Majority Element (Boyer-Moore)**	High	Easy	NQT/Digital	YES	*Boyer-Moore voting*
31	**Find the Element that Appears Once (others appear twice)**	High	Easy	NQT	YES	*XOR all elements*
32	**Find the Element that Appears Once (others appear thrice)**	Medium	Medium	Digital	YES	*Bit counting*
33	**Product of Array Except Self**	Medium	Medium	Digital/Prime	YES	*Left and right product arrays*
34	**Sort Array of 0s 1s 2s (Dutch National Flag)**	Very High	Easy	NQT/Digital	YES	*3-pointer in-place*
35	**Find Kth Smallest / Largest Element**	High	Medium	Digital/Prime	YES	*QuickSelect or heap*
36	**Minimum Platforms for Train Schedule**	High	Medium	Digital	YES	*Sort arrivals & departures*
37	**Maximum Sum of Non-Adjacent Elements**	High	Medium	Digital	YES	*DP House Robber*
38	**Chocolate Distribution Problem**	Medium	Easy	NQT	YES	*Sort, sliding window size m*
39	**Minimum Number of Jumps to Reach End**	High	Medium	Digital/Prime	YES	*Greedy O(n)*

#	Question	Freq	Diff	Round	Rep?	Description
40	**Next Permutation**	Medium	Medium	Digital	YES	*STL logic manual*
41	**Find First and Last Position of Element in Sorted Array**	Medium	Easy	Digital	YES	*Two binary searches*
42	**Count Occurrences of Element in Sorted Array**	Medium	Easy	NQT	YES	*Upper – lower bound*
43	**Wave Array**	Medium	Easy	NQT	YES	*Swap adjacent pairs*
44	**Segregate Even and Odd Numbers**	Medium	Easy	NQT	YES	*Two-pointer*
45	**Find the Median of Two Sorted Arrays**	Medium	Hard	Prime	YES	*Binary search O(log min(m,n))*
46	**Maximum Length Bitonic Subarray**	Low	Medium	Digital	no	*DP inc + dec arrays*
47	**Count Total Set Bits in All Numbers 1 to N**	Low	Medium	Digital	no	*Bit pattern grouping*
48	**Longest Mountain Subarray**	Low	Medium	Prime	no	*Two pass expand*

Sorting (14 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Bubble Sort Implementation**	Very High	Easy	NQT	YES	*O(n^2) stable*
2	**Selection Sort Implementation**	Very High	Easy	NQT	YES	*O(n^2) unstable*
3	**Insertion Sort Implementation**	Very High	Easy	NQT	YES	*O(n^2) stable adaptive*
4	**Merge Sort Implementation**	Very High	Medium	NQT/Digital	YES	*O(n log n) stable divide & conquer*
5	**Quick Sort Implementation**	Very High	Medium	NQT/Digital	YES	*O(n log n) avg partition*
6	**Heap Sort Implementation**	High	Medium	Digital	YES	*O(n log n) using max-heap*
7	**Counting Sort**	Medium	Easy	NQT/Digital	YES	*O(n+k) for small range*
8	**Radix Sort**	Medium	Medium	Digital	YES	*Digit-by-digit counting sort*
9	**Shell Sort**	Low	Medium	NQT	no	*Gap sequence insertion*
10	**Kth Largest Element using Quick Select**	High	Medium	Digital/Prime	YES	*Avg O(n) partition-based*

#	Question	Freq	Diff	Round	Rep?	Description
11	**Sort Characters by Frequency**	Medium	Medium	Digital	YES	*Char frequency + sort*
12	**Sort Array by Parity**	Medium	Easy	NQT	YES	*Even elements first*
13	**Custom Comparator Sorting – Sort by second element of pair**	Medium	Easy	Digital	YES	*Lambda / custom compare*
14	**Pancake Sorting**	Low	Medium	Prime	no	*Flip operations*

Binary Search (12 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Binary Search – Standard**	Very High	Easy	NQT	YES	*O(log n) iterative*
2	**Binary Search on Answer – Minimum Max Subarray**	High	Hard	Prime	YES	*BS on answer space*
3	**Floor and Ceiling in Sorted Array**	High	Easy	NQT/Digital	YES	*Lower_bound / upper_bound logic*
4	**Square Root using Binary Search**	High	Easy	NQT/Digital	YES	*BS on 1 to n/2*
5	**Peak Element in Array**	High	Medium	Digital	YES	*Recursive BS*
6	**Find Rotation Count in Rotated Sorted Array**	Medium	Medium	Digital	YES	*BS on pivot*
7	**Median in Row-wise Sorted Matrix**	Medium	Hard	Prime	YES	*BS + row counts*
8	**Allocate Minimum Number of Pages**	High	Hard	Digital/Prime	YES	*BS on answer*
9	**Aggressive Cows (Maximum Minimum Distance)**	High	Hard	Digital/Prime	YES	*BS on answer + greedy check*
10	**Find Position of Element in Infinite Sorted Array**	Low	Medium	Digital	no	*Exponential search + BS*
11	**Kth Smallest Element in Sorted Matrix**	Medium	Hard	Prime	YES	*BS on value space*
12	**Search in Nearly Sorted Array**	Low	Easy	NQT	no	*Check mid-1, mid, mid+1*

Strings (30 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Reverse a String**	Very High	Easy	NQT	YES	*Two-pointer*

#	Question	Freq	Diff	Round	Rep?	Description
2	**Check Palindrome**	Very High	Easy	NQT	YES	*Two-pointer check*
3	**Check Anagram**	Very High	Easy	NQT/Digital	YES	*Frequency map or sort*
4	**Count Vowels and Consonants**	High	Easy	NQT	YES	*Iterate, classify*
5	**Remove Duplicate Characters**	High	Easy	NQT/Digital	YES	*Set tracking*
6	**First Non-Repeating Character**	Very High	Easy	NQT/Digital	YES	*LinkedHashMap frequency*
7	**Longest Palindromic Substring**	High	Medium	Digital/Prime	YES	*Expand around center*
8	**Longest Common Prefix**	High	Easy	NQT/Digital	YES	*Horizontal scanning*
9	**String Rotation Check**	High	Easy	NQT	YES	*Concatenate and search*
10	**Count and Say Sequence**	Medium	Easy	NQT	YES	*RLE encoding*
11	**Roman to Integer**	High	Easy	NQT/Digital	YES	*Map lookup*
12	**Integer to Roman**	Medium	Easy	Digital	YES	*Greedy subtraction*
13	**Valid Parentheses**	Very High	Easy	NQT/Digital	YES	*Stack push/pop*
14	**Longest Valid Parentheses**	Medium	Hard	Prime	YES	*Stack with indices or DP*
15	**Minimum Window Substring**	Medium	Hard	Prime	YES	*Sliding window + freq map*
16	**Group Anagrams Together**	High	Medium	Digital	YES	*Sort each string as key*
17	**Rabin-Karp String Search**	Medium	Medium	Digital	no	*Rolling hash*
18	**KMP String Search**	Medium	Medium	Digital/Prime	YES	*Prefix function*
19	**Z Algorithm**	Low	Medium	Prime	no	*Z-array pattern match*
20	**Compress String (Run Length Encoding)**	High	Easy	NQT	YES	*Count consecutive chars*
21	**Reverse Words in a String**	High	Easy	NQT/Digital	YES	*Split and reverse*
22	**Isomorphic Strings**	Medium	Easy	NQT/Digital	YES	*Two-way character mapping*
23	**Wildcard Pattern Matching**	Medium	Hard	Prime	YES	*DP table*

#	Question	Freq	Diff	Round	Rep?	Description
24	**Regular Expression Matching**	Low	Hard	Prime	no	*DP with . and **
25	**Permutations of a String**	High	Medium	Digital	YES	*Backtracking*
26	**Word Break Problem**	High	Medium	Digital/Prime	YES	*DP + dictionary set*
27	**Decode Ways**	Medium	Medium	Digital	YES	*DP decoding*
28	**Longest Substring Without Repeating Characters**	Very High	Medium	Digital/Prime	YES	*Sliding window + set*
29	**Count Substrings with K Distinct Characters**	Medium	Medium	Digital	YES	*Sliding window*
30	**Print All Subsequences of a String**	Medium	Medium	Digital	YES	*Recursion / bitmask*

Hashing (10 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Two Sum using HashMap**	Very High	Easy	NQT/Digital	YES	*O(n) complement lookup*
2	**Find All Pairs with Zero Sum**	High	Easy	NQT/Digital	YES	*Prefix sum map*
3	**Subarray with Given Sum (negative numbers)**	High	Medium	Digital	YES	*Prefix sum HashMap*
4	**Count Distinct Elements in Every Window**	Medium	Medium	Digital	YES	*Sliding window + freq map*
5	**Find Longest Consecutive Sequence**	High	Medium	Digital/Prime	YES	*HashSet, check start of seq*
6	**4-Sum Problem**	Medium	Medium	Digital	YES	*HashMap or sort+2ptr*
7	**Check if Array is Subset of Another**	Medium	Easy	NQT	YES	*HashSet membership*
8	**First Repeating Element**	High	Easy	NQT	YES	*HashMap track first occurrence*
9	**Frequency of Each Element**	High	Easy	NQT	YES	*Basic freq map*
10	**Group Integers with Same Digit Sum**	Low	Medium	Digital	no	*Map digit sum -> list*

Linked List (19 questions)

#	Question	Freq	Diff	Round	Rep ?	Description
1	**Reverse a Linked List**	Very High	Easy	NQT/Digital	YES	*Iterative 3-pointer*
2	**Detect Loop in Linked List**	Very High	Easy	NQT/Digital	YES	*Floyd's cycle detection*
3	**Find Start of Loop**	High	Medium	Digital	YES	*Floyd's + meeting point math*
4	**Find Middle of Linked List**	Very High	Easy	NQT	YES	*Slow-fast pointer*
5	**Remove Nth Node from End**	High	Easy	NQT/Digital	YES	*Two-pointer gap of N*
6	**Merge Two Sorted Linked Lists**	Very High	Easy	NQT/Digital	YES	*Recursive or iterative*
7	**Merge K Sorted Linked Lists**	Medium	Hard	Prime	YES	*Min-heap approach*
8	**Check Palindrome Linked List**	High	Medium	NQT/Digital	YES	*Find mid, reverse second half*
9	**Delete Node Without Head Pointer**	High	Easy	Digital	YES	*Copy next data, skip next*
10	**Flatten a Multilevel Linked List**	Medium	Medium	Digital	YES	*Recursion / stack*
11	**Add Two Numbers as Linked Lists**	High	Medium	NQT/Digital	YES	*Digit-by-digit carry*
12	**Intersection Point of Two Linked Lists**	High	Medium	NQT/Digital	YES	*Length difference or two-pointer*
13	**Reverse Linked List in Groups of K**	High	Medium	Digital/Prime	YES	*Recursive group reversal*
14	**Sort Linked List (Merge Sort)**	High	Medium	Digital	YES	*Find mid, split, merge*
15	**Clone Linked List with Random Pointer**	Medium	Hard	Prime	YES	*HashMap node mapping*
16	**Segregate Even and Odd Nodes**	Medium	Easy	NQT	YES	*Two separate lists*
17	**Rotate Linked List by K**	Medium	Medium	Digital	YES	*Find (n-k)th node, reconnect*
18	**Delete All Occurrences of a Key**	Medium	Easy	NQT	YES	*Traverse and skip*
19	**LRU Cache Implementation**	High	Hard	Digital/Prime	YES	*DLL + HashMap*

Stack (17 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Implement Stack using Array**	Very High	Easy	NQT	YES	*Array-backed push/pop*
2	**Implement Stack using Queue**	Medium	Medium	Digital	YES	*Two queues trick*
3	**Implement Queue using Stack**	Medium	Medium	Digital	YES	*Two stacks trick*
4	**Valid Parentheses Check**	Very High	Easy	NQT/Digital	YES	*Push open, pop on close*
5	**Next Greater Element**	Very High	Medium	NQT/Digital	YES	*Monotonic stack O(n)*
6	**Previous Smaller Element**	High	Medium	Digital	YES	*Monotonic stack O(n)*
7	**Stock Span Problem**	High	Medium	NQT/Digital	YES	*Stack of (price, span) pairs*
8	**Largest Rectangle in Histogram**	High	Hard	Digital/Prime	YES	*Monotonic stack left/right extents*
9	**Maximum Area Rectangle in Binary Matrix**	High	Hard	Prime	YES	*Histogram row by row*
10	**Celebrity Problem**	High	Medium	Digital	YES	*Stack elimination*
11	**Sort a Stack (recursion)**	High	Medium	NQT/Digital	YES	*Recursive insert in sorted order*
12	**Reverse a Stack (recursion)**	Medium	Medium	Digital	YES	*Insert at bottom recursively*
13	**Evaluate Postfix Expression**	High	Easy	NQT/Digital	YES	*Operand push, operator pop-calc*
14	**Infix to Postfix Conversion**	High	Medium	NQT/Digital	YES	*Operator precedence stack*
15	**Infix to Prefix Conversion**	Medium	Medium	Digital	YES	*Reverse, to-postfix, reverse*
16	**Min Stack (O(1) getMin)**	High	Medium	Digital/Prime	YES	*Auxiliary min stack or encoding*
17	**Decode String (nested brackets)**	Medium	Medium	Digital	YES	*Count and string stacks*

Queue (8 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Implement Queue using Array**	Very High	Easy	NQT	YES	*Circular array*
2	**Implement Circular Queue**	High	Medium	Digital	YES	*Modulo indexing*

#	Question	Freq	Diff	Round	Rep?	Description
3	**Sliding Window Maximum using Deque**	High	Hard	Digital/Prime	YES	*Monotonic deque front is max*
4	**First Non-Repeating Character in Stream**	High	Medium	Digital	YES	*Queue + freq map*
5	**BFS of Graph using Queue**	Very High	Medium	NQT/Digital	YES	*Level-order traversal*
6	**Level Order Traversal of Tree**	Very High	Easy	NQT/Digital	YES	*BFS with queue*
7	**Rotten Oranges – Multi-source BFS**	High	Medium	Digital/Prime	YES	*BFS from all rotten at once*
8	**Interleave First and Second Half of Queue**	Medium	Medium	Digital	YES	*Split + interleave*

Trees (33 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Inorder Traversal (recursive & iterative)**	Very High	Easy	NQT/Digital	YES	*Left-Root-Right*
2	**Preorder Traversal**	Very High	Easy	NQT/Digital	YES	*Root-Left-Right*
3	**Postorder Traversal**	Very High	Easy	NQT/Digital	YES	*Left-Right-Root*
4	**Level Order Traversal**	Very High	Easy	NQT/Digital	YES	*BFS with queue*
5	**Height / Depth of Binary Tree**	Very High	Easy	NQT/Digital	YES	*max(left,right)+1*
6	**Diameter of Binary Tree**	High	Medium	Digital/Prime	YES	*Max path through root*
7	**Check if Binary Tree is Balanced**	High	Medium	Digital	YES	*Height check each node*
8	**Mirror / Invert Binary Tree**	High	Easy	NQT/Digital	YES	*Swap children recursively*
9	**Check if Two Trees are Identical**	High	Easy	NQT	YES	*Recursive structure & val*
10	**Lowest Common Ancestor (BST)**	High	Easy	NQT/Digital	YES	*Navigate by value*
11	**Lowest Common Ancestor (Binary Tree)**	High	Medium	Digital/Prime	YES	*Recursive search both subtrees*
12	**Path from Root to Node**	High	Medium	Digital	YES	*DFS backtracking*
13	**Zigzag Level Order Traversal**	High	Medium	Digital	YES	*Level BFS, alternate direction*

#	Question	Freq	Diff	Round	Rep?	Description
14	**Vertical Order Traversal**	High	Medium	Digital/Prime	YES	*BFS + column map*
15	**Top View of Binary Tree**	High	Medium	Digital	YES	*BFS horizontal distance*
16	**Bottom View of Binary Tree**	High	Medium	Digital	YES	*BFS horizontal distance last level*
17	**Left View / Right View of Binary Tree**	High	Easy	NQT/Digital	YES	*Level-first node tracking*
18	**BST – Insert, Delete, Search**	Very High	Easy	NQT/Digital	YES	*Recursive BST ops*
19	**BST – Check if Valid BST**	High	Medium	Digital	YES	*Inorder is sorted or range check*
20	**BST – Kth Smallest Element**	High	Medium	Digital	YES	*Inorder traversal count*
21	**BST – Convert to Sorted DLL**	Medium	Hard	Prime	YES	*In-place inorder link change*
22	**Serialize and Deserialize Binary Tree**	Medium	Hard	Prime	YES	*Preorder with null markers*
23	**Max Path Sum in Binary Tree**	High	Hard	Prime	YES	*Recursive left+right+root*
24	**Boundary Traversal of Binary Tree**	Medium	Medium	Digital	YES	*Left boundary + leaves + right*
25	**Count Nodes in Complete Binary Tree**	Medium	Medium	Digital	YES	*Binary search on levels*
26	**Flatten Binary Tree to Linked List**	Medium	Medium	Digital	YES	*Reverse postorder threading*
27	**Construct Tree from Inorder and Preorder**	High	Medium	Digital/Prime	YES	*Preorder root, split inorder*
28	**Construct Tree from Inorder and Postorder**	Medium	Medium	Digital	YES	*Postorder root, split inorder*
29	**Segment Tree – Build and Query**	Medium	Hard	Prime	YES	*Range sum/min/max queries*
30	**Fenwick Tree / BIT**	Low	Hard	Prime	no	*Prefix sum with bit tricks*
31	**Trie – Insert and Search**	High	Medium	Digital/Prime	YES	*26-child node structure*
32	**Trie – Longest Prefix Matching**	Medium	Medium	Prime	YES	*Walk trie as far as possible*
33	**Check if Tree is Symmetric**	Medium	Easy	NQT	YES	*Mirror check recursion*

Graphs (25 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**BFS of Graph**	Very High	Easy	NQT/Digital	YES	*Queue, visited array*
2	**DFS of Graph**	Very High	Easy	NQT/Digital	YES	*Stack / recursion*
3	**Detect Cycle in Undirected Graph (BFS/DFS)**	Very High	Medium	Digital/Prime	YES	*Parent tracking in BFS*
4	**Detect Cycle in Directed Graph (DFS)**	Very High	Medium	Digital/Prime	YES	*Recursion stack flag*
5	**Topological Sort (DFS)**	High	Medium	Digital/Prime	YES	*DFS finish order reverse*
6	**Topological Sort (Kahn's BFS)**	High	Medium	Digital/Prime	YES	*In-degree queue*
7	**Shortest Path – Dijkstra**	High	Medium	Digital/Prime	YES	*Min-heap priority queue*
8	**Shortest Path – Bellman-Ford**	Medium	Medium	Digital/Prime	YES	*Relax all edges V-1 times*
9	**Shortest Path – Floyd-Warshall (All Pairs)**	Medium	Medium	Digital	YES	*DP through intermediate nodes*
10	**Minimum Spanning Tree – Prim's**	High	Medium	Digital/Prime	YES	*Greedy grow from source*
11	**Minimum Spanning Tree – Kruskal's**	High	Medium	Digital/Prime	YES	*Sort edges + Union-Find*
12	**Number of Islands (BFS/DFS on matrix)**	Very High	Medium	NQT/Digital	YES	*Mark visited cells*
13	**Flood Fill Algorithm**	High	Medium	NQT/Digital	YES	*DFS color replacement*
14	**Word Ladder – BFS shortest transform**	High	Hard	Prime	YES	*BFS word graph*
15	**Bipartite Check**	Medium	Medium	Digital	YES	*2-color BFS/DFS*
16	**Strongly Connected Components – Kosaraju**	Medium	Hard	Prime	YES	*Two DFS passes*
17	**Strongly Connected Components – Tarjan**	Low	Hard	Prime	no	*Single DFS low-link*
18	**Bridge and Articulation Point**	Low	Hard	Prime	no	*DFS low values*
19	**Clone a Graph**	Medium	Medium	Digital	YES	*BFS + HashMap old->new*
20	**Alien Dictionary (Topological Sort)**	Medium	Hard	Prime	YES	*Build char graph from ordering*
21	**Course Schedule I & II**	High	Medium	Digital/Prime	YES	*Cycle detect / topo sort*
22	**Number of Connected Components**	High	Medium	NQT/Digital	YES	*BFS/DFS count starts*

#	Question	Freq	Diff	Round	Rep?	Description
23	**Grid Shortest Path (0-1 BFS)**	Medium	Medium	Digital	YES	*Deque-based 0-1 BFS*
24	**Snakes and Ladders – Shortest Path**	Medium	Medium	Digital	YES	*BFS on board cells*
25	**Is Graph a Tree (no cycle + connected)**	Medium	Easy	NQT/Digital	YES	*V-1 edges + connected*

Recursion (17 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Fibonacci (recursive, memoized, iterative)**	Very High	Easy	NQT	YES	*Base cases 0,1*
2	**Factorial**	Very High	Easy	NQT	YES	*n * fact(n-1)*
3	**Tower of Hanoi**	High	Medium	NQT/Digital	YES	*3-move recursive*
4	**Generate All Subsets / Power Set**	High	Medium	Digital/Prime	YES	*Include/exclude recursion*
5	**Generate All Permutations**	High	Medium	Digital/Prime	YES	*Swap backtrack*
6	**N-Queens Problem**	High	Hard	Digital/Prime	YES	*Place queens row by row*
7	**Sudoku Solver**	Medium	Hard	Prime	YES	*Backtrack cell by cell*
8	**Rat in a Maze**	High	Medium	Digital	YES	*DFS 4 directions backtrack*
9	**Print All Paths from Source to Destination**	High	Medium	Digital	YES	*DFS path tracking*
10	**Word Search in Matrix**	High	Medium	Digital/Prime	YES	*DFS from each cell*
11	**Combination Sum I (target sum, reuse)**	High	Medium	Digital/Prime	YES	*Backtrack with start index*
12	**Combination Sum II (no reuse)**	Medium	Medium	Digital	YES	*Skip duplicates*
13	**Subset Sum Problem**	High	Medium	Digital/Prime	YES	*Include/exclude DP or backtrack*
14	**Print K-th Permutation**	Medium	Hard	Prime	YES	*Factorial number system*
15	**Generate Parentheses**	High	Medium	Digital	YES	*Backtrack open/close counts*
16	**Partition Equal Subset Sum**	High	Medium	Digital/Prime	YES	*0/1 knapsack DP*

#	Question	Freq	Diff	Round	Rep?	Description
17	**Flood Fill Recursive**	High	Easy	NQT/Digital	YES	*4-directional DFS*

Dynamic Programming (36 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**0/1 Knapsack**	Very High	Medium	Digital/Prime	YES	*dp[i][w] include/exclude*
2	**Unbounded Knapsack**	High	Medium	Digital/Prime	YES	*Reuse items allowed*
3	**Longest Common Subsequence (LCS)**	Very High	Medium	Digital/Prime	YES	*dp[i][j] 2D table*
4	**Longest Increasing Subsequence (LIS)**	Very High	Medium	Digital/Prime	YES	*dp O(n^2) or patience sort O(n log n)*
5	**Edit Distance (Levenshtein)**	High	Medium	Digital/Prime	YES	*Insert/delete/replace cost*
6	**Coin Change – Minimum Coins**	Very High	Medium	NQT/Digital/Prime	YES	*dp[amount] bottom-up*
7	**Coin Change – Count Ways**	High	Medium	Digital/Prime	YES	*Unbounded knapsack variant*
8	**Matrix Chain Multiplication**	High	Hard	Prime	YES	*dp[i][j] min cost*
9	**Egg Drop Problem**	Medium	Hard	Prime	YES	*dp[eggs][floors]*
10	**Word Break Problem**	High	Medium	Digital/Prime	YES	*dp[i] prefix reachable*
11	**Palindrome Partitioning – Minimum Cuts**	Medium	Hard	Prime	YES	*DP + palindrome check*
12	**Palindrome Partitioning – All Partitions**	Medium	Hard	Prime	YES	*Backtrack + palindrome*
13	**Rod Cutting Problem**	High	Medium	Digital/Prime	YES	*Unbounded knapsack analogy*
14	**Maximum Sum Increasing Subsequence**	High	Medium	Digital	YES	*LIS variant with sum*
15	**Longest Bitonic Subsequence**	Medium	Medium	Digital	YES	*LIS forward + backward*
16	**Print LCS**	High	Medium	Digital/Prime	YES	*Traceback dp table*
17	**Shortest Common Supersequence**	Medium	Medium	Digital	YES	*SCS length = m+n-LCS*
18	**Distinct Subsequences**	Medium	Hard	Prime	YES	*dp[i][j] count matches*

#	Question	Freq	Diff	Round	Rep?	Description
19	**Wildcard Matching DP**	Medium	Hard	Prime	YES	*dp with * and ? handling*
20	**Interleaving Strings**	Medium	Hard	Prime	YES	*dp[i][j] two-string interleave*
21	**Minimum Path Sum in Grid**	High	Medium	Digital/Prime	YES	*dp[i][j] = min from top/left*
22	**Unique Paths in Grid**	High	Easy	Digital	YES	*Combinatorial or dp*
23	**Unique Paths with Obstacles**	Medium	Medium	Digital	YES	*dp skip obstacle cells*
24	**Maximum Square of 1s in Matrix**	High	Medium	Digital/Prime	YES	*dp[i][j] min of neighbours+1*
25	**Largest Rectangle in Histogram DP**	Medium	Hard	Prime	YES	*Stack or DP left/right extents*
26	**Burst Balloons**	Low	Hard	Prime	no	*Interval DP*
27	**Partition Array for Maximum Sum**	Low	Medium	Prime	no	*Interval DP*
28	**House Robber I**	High	Easy	Digital	YES	*dp max non-adjacent*
29	**House Robber II (circular)**	Medium	Medium	Digital	YES	*Two runs: 0..n-2, 1..n-1*
30	**Jump Game I (can reach end?)**	High	Medium	Digital	YES	*Greedy max reach*
31	**Jump Game II (minimum jumps)**	High	Medium	Digital/Prime	YES	*Greedy level BFS*
32	**Palindromic Substrings Count**	Medium	Medium	Digital	YES	*Expand or DP*
33	**Longest Palindromic Subsequence**	Medium	Medium	Digital/Prime	YES	*LCS of s and reverse(s)*
34	**Stock Buy Sell with Cooldown**	Low	Medium	Prime	no	*3-state DP*
35	**Stock Buy Sell with Transaction Fee**	Low	Medium	Prime	no	*2-state DP*
36	**Decode Ways**	Medium	Medium	Digital	YES	*dp[i] one/two digit decode*

Bit Manipulation (12 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Count Set Bits in Integer**	High	Easy	NQT/Digital	YES	*Brian Kernighan n&(n-1)*

#	Question	Freq	Diff	Round	Rep?	Description
2	**Check if Power of 2**	High	Easy	NQT/Digital	YES	*n & (n-1) == 0*
3	**Find XOR of All Numbers from 1 to N**	High	Easy	NQT	YES	*Pattern every 4 numbers*
4	**Swap Two Numbers without Temp**	High	Easy	NQT	YES	*XOR swap*
5	**Find the Two Non-Repeating Elements**	High	Medium	Digital	YES	*XOR partition by set bit*
6	**Reverse Bits of Integer**	Medium	Easy	NQT/Digital	YES	*Shift and OR*
7	**Find Position of Only Set Bit**	Medium	Easy	NQT	YES	*log2 or bit scan*
8	**Power Set using Bits**	Medium	Medium	Digital	YES	*Bitmask 0 to 2^n-1*
9	**Hamming Distance**	Medium	Easy	NQT/Digital	YES	*Count bits in XOR*
10	**Add Two Numbers without + Operator**	Medium	Medium	Digital	YES	*XOR sum + AND carry*
11	**Multiply by Power of 2**	Low	Easy	NQT	no	*Left shift*
12	**Divide by Power of 2**	Low	Easy	NQT	no	*Right shift*

Matrix (12 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Rotate Matrix 90 Degrees**	High	Medium	NQT/Digital	YES	*Transpose then reverse rows*
2	**Spiral Matrix Traversal**	High	Medium	NQT/Digital	YES	*Layer-by-layer boundary*
3	**Search in Row-Column Sorted Matrix**	High	Medium	Digital	YES	*Start top-right, O(m+n)*
4	**Set Matrix Zeroes**	High	Medium	Digital	YES	*Mark rows/cols first*
5	**Transpose of Matrix**	High	Easy	NQT	YES	*Swap a[i][j] and a[j][i]*
6	**Scalar Multiplication of Matrix**	Medium	Easy	NQT	YES	*Element-wise multiply*
7	**Matrix Multiplication**	High	Medium	NQT/Digital	YES	*O(n^3) naive*
8	**Diagonal Sum of Matrix**	Medium	Easy	NQT	YES	*Track primary+secondary*

#	Question	Freq	Diff	Round	Rep?	Description
9	**Print Matrix in Zigzag**	Medium	Medium	Digital	YES	*Row-by-row alternating*
10	**Find Maximum in each Row**	Medium	Easy	NQT	YES	*Simple row scan*
11	**Count Islands in Binary Matrix**	High	Medium	Digital	YES	*DFS mark visited*
12	**Minimum Cost Path in Matrix**	High	Medium	Digital/Prime	YES	*DP dp[i][j]*

Number Theory (20 questions)

#	Question	Freq	Diff	Round	Rep?	Description
1	**Check Prime Number**	Very High	Easy	NQT	YES	*Trial division up to sqrt*
2	**Sieve of Eratosthenes**	High	Easy	NQT/Digital	YES	*Boolean array mark composites*
3	**GCD and LCM**	Very High	Easy	NQT	YES	*Euclidean algorithm*
4	**Check Armstrong Number**	Very High	Easy	NQT	YES	*Sum of digit^n == number*
5	**Check Perfect Number**	High	Easy	NQT	YES	*Sum of divisors == number*
6	**Check Strong Number**	High	Easy	NQT	YES	*Sum of digit factorials == number*
7	**Fibonacci Series (many methods)**	Very High	Easy	NQT	YES	*Iterative or matrix exponentiation*
8	**Power Function – Fast Exponentiation**	High	Medium	NQT/Digital	YES	*Binary exponentiation O(log n)*
9	**Modular Exponentiation**	High	Medium	Digital/Prime	YES	*(base^exp) % mod*
10	**Extended Euclidean Algorithm**	Low	Medium	Prime	no	*ax + by = gcd(a,b)*
11	**Check Palindrome Number**	Very High	Easy	NQT	YES	*Reverse digits compare*
12	**Digit Sum / Digital Root**	High	Easy	NQT	YES	*Sum digits repeatedly*
13	**Count Digits in Number**	High	Easy	NQT	YES	*log10 + 1 or loop*
14	**Reverse Digits of Number**	High	Easy	NQT	YES	*Mod 10 extract, build reverse*
15	**Check Automorphic Number**	Medium	Easy	NQT	no	*n^2 ends with n*

#	Question	Freq	Diff	Round	Rep?	Description
16	**Find All Divisors of N**	High	Easy	NQT/Digital	YES	*Loop to sqrt(n)*
17	**Catalan Number**	Medium	Medium	Digital	YES	*C(2n,n)/(n+1) or DP*
18	**Pascal's Triangle**	High	Easy	NQT	YES	*Row-by-row DP*
19	**Check Neon Number**	Medium	Easy	NQT	no	*Sum of digits of square == number*
20	**Ugly Numbers**	Medium	Medium	Digital	YES	*DP track multiples of 2,3,5*

TOP 25 MOST FREQUENT DSA QUESTIONS

#	Question	Topic	Freq	Diff	Round
1	**Find Second Largest Element**	Arrays	Very High	Easy	NQT/Digital
2	**Find Missing Number (1 to N)**	Arrays	Very High	Easy	NQT
3	**Kadane's Algorithm – Maximum Subarray Sum**	Arrays	Very High	Medium	Digital/Prime
4	**Find Pair with Given Sum (Two Sum)**	Arrays	Very High	Easy	NQT/Digital
5	**Best Time to Buy and Sell Stock**	Arrays	Very High	Easy	Digital/Prime
6	**Sort Array of 0s 1s 2s (Dutch National Flag)**	Arrays	Very High	Easy	NQT/Digital
7	**Bubble Sort Implementation**	Sorting	Very High	Easy	NQT
8	**Selection Sort Implementation**	Sorting	Very High	Easy	NQT
9	**Insertion Sort Implementation**	Sorting	Very High	Easy	NQT
10	**Merge Sort Implementation**	Sorting	Very High	Medium	NQT/Digital
11	**Quick Sort Implementation**	Sorting	Very High	Medium	NQT/Digital
12	**Binary Search – Standard**	Binary Search	Very High	Easy	NQT
13	**Reverse a String**	Strings	Very High	Easy	NQT
14	**Check Palindrome**	Strings	Very High	Easy	NQT
15	**Check Anagram**	Strings	Very High	Easy	NQT/Digital
16	**First Non-Repeating Character**	Strings	Very High	Easy	NQT/Digital
17	**Valid Parentheses**	Strings	Very High	Easy	NQT/Digital
18	**Longest Substring Without Repeating Characters**	Strings	Very High	Medium	Digital/Prime
19	**Two Sum using HashMap**	Hashing	Very High	Easy	NQT/Digital
20	**Reverse a Linked List**	Linked List	Very High	Easy	NQT/Digital
21	**Detect Loop in Linked List**	Linked List	Very High	Easy	NQT/Digital
22	**Find Middle of Linked List**	Linked List	Very High	Easy	NQT
23	**Merge Two Sorted Linked Lists**	Linked List	Very High	Easy	NQT/Digital

#	Question	Topic	Freq	Diff	Round
24	**Implement Stack using Array**	Stack	Very High	Easy	NQT
25	**Valid Parentheses Check**	Stack	Very High	Easy	NQT/Digital

TOP 50 MOST FREQUENT DSA QUESTIONS

#	Question	Topic	Freq	Diff	Round
1	**Find Second Largest Element**	Arrays	Very High	Easy	NQT/Digital
2	**Find Missing Number (1 to N)**	Arrays	Very High	Easy	NQT
3	**Kadane's Algorithm – Maximum Subarray Sum**	Arrays	Very High	Medium	Digital/Prime
4	**Find Pair with Given Sum (Two Sum)**	Arrays	Very High	Easy	NQT/Digital
5	**Best Time to Buy and Sell Stock**	Arrays	Very High	Easy	Digital/Prime
6	**Sort Array of 0s 1s 2s (Dutch National Flag)**	Arrays	Very High	Easy	NQT/Digital
7	**Bubble Sort Implementation**	Sorting	Very High	Easy	NQT
8	**Selection Sort Implementation**	Sorting	Very High	Easy	NQT
9	**Insertion Sort Implementation**	Sorting	Very High	Easy	NQT
10	**Merge Sort Implementation**	Sorting	Very High	Medium	NQT/Digital
11	**Quick Sort Implementation**	Sorting	Very High	Medium	NQT/Digital
12	**Binary Search – Standard**	Binary Search	Very High	Easy	NQT
13	**Reverse a String**	Strings	Very High	Easy	NQT
14	**Check Palindrome**	Strings	Very High	Easy	NQT
15	**Check Anagram**	Strings	Very High	Easy	NQT/Digital
16	**First Non-Repeating Character**	Strings	Very High	Easy	NQT/Digital
17	**Valid Parentheses**	Strings	Very High	Easy	NQT/Digital
18	**Longest Substring Without Repeating Characters**	Strings	Very High	Medium	Digital/Prime
19	**Two Sum using HashMap**	Hashing	Very High	Easy	NQT/Digital
20	**Reverse a Linked List**	Linked List	Very High	Easy	NQT/Digital

#	Question	Topic	Freq	Diff	Round
21	**Detect Loop in Linked List**	Linked List	Very High	Easy	NQT/Digital
22	**Find Middle of Linked List**	Linked List	Very High	Easy	NQT
23	**Merge Two Sorted Linked Lists**	Linked List	Very High	Easy	NQT/Digital
24	**Implement Stack using Array**	Stack	Very High	Easy	NQT
25	**Valid Parentheses Check**	Stack	Very High	Easy	NQT/Digital
26	**Next Greater Element**	Stack	Very High	Medium	NQT/Digital
27	**Implement Queue using Array**	Queue	Very High	Easy	NQT
28	**BFS of Graph using Queue**	Queue	Very High	Medium	NQT/Digital
29	**Level Order Traversal of Tree**	Queue	Very High	Easy	NQT/Digital
30	**Inorder Traversal (recursive & iterative)**	Trees	Very High	Easy	NQT/Digital
31	**Preorder Traversal**	Trees	Very High	Easy	NQT/Digital
32	**Postorder Traversal**	Trees	Very High	Easy	NQT/Digital
33	**Level Order Traversal**	Trees	Very High	Easy	NQT/Digital
34	**Height / Depth of Binary Tree**	Trees	Very High	Easy	NQT/Digital
35	**BST – Insert, Delete, Search**	Trees	Very High	Easy	NQT/Digital
36	**BFS of Graph**	Graphs	Very High	Easy	NQT/Digital
37	**DFS of Graph**	Graphs	Very High	Easy	NQT/Digital
38	**Detect Cycle in Undirected Graph (BFS/DFS)**	Graphs	Very High	Medium	Digital/Prime
39	**Detect Cycle in Directed Graph (DFS)**	Graphs	Very High	Medium	Digital/Prime
40	**Number of Islands (BFS/DFS on matrix)**	Graphs	Very High	Medium	NQT/Digital
41	**Fibonacci (recursive, memoized, iterative)**	Recursion	Very High	Easy	NQT
42	**Factorial**	Recursion	Very High	Easy	NQT

#	Question	Topic	Freq	Diff	Round
43	**0/1 Knapsack**	Dynamic Programming	Very High	Medium	Digital/Prime
44	**Longest Common Subsequence (LCS)**	Dynamic Programming	Very High	Medium	Digital/Prime
45	**Longest Increasing Subsequence (LIS)**	Dynamic Programming	Very High	Medium	Digital/Prime
46	**Coin Change – Minimum Coins**	Dynamic Programming	Very High	Medium	NQT/Digital/Prime
47	**Check Prime Number**	Number Theory	Very High	Easy	NQT
48	**GCD and LCM**	Number Theory	Very High	Easy	NQT
49	**Check Armstrong Number**	Number Theory	Very High	Easy	NQT
50	**Fibonacci Series (many methods)**	Number Theory	Very High	Easy	NQT

TOP 100 MOST FREQUENT DSA QUESTIONS

#	Question	Topic	Freq	Diff	Round
1	**Find Second Largest Element**	Arrays	Very High	Easy	NQT/Digital
2	**Find Missing Number (1 to N)**	Arrays	Very High	Easy	NQT
3	**Kadane's Algorithm – Maximum Subarray Sum**	Arrays	Very High	Medium	Digital/Prime
4	**Find Pair with Given Sum (Two Sum)**	Arrays	Very High	Easy	NQT/Digital
5	**Best Time to Buy and Sell Stock**	Arrays	Very High	Easy	Digital/Prime
6	**Sort Array of 0s 1s 2s (Dutch National Flag)**	Arrays	Very High	Easy	NQT/Digital
7	**Bubble Sort Implementation**	Sorting	Very High	Easy	NQT
8	**Selection Sort Implementation**	Sorting	Very High	Easy	NQT
9	**Insertion Sort Implementation**	Sorting	Very High	Easy	NQT
10	**Merge Sort Implementation**	Sorting	Very High	Medium	NQT/Digital
11	**Quick Sort Implementation**	Sorting	Very High	Medium	NQT/Digital
12	**Binary Search – Standard**	Binary Search	Very High	Easy	NQT
13	**Reverse a String**	Strings	Very High	Easy	NQT
14	**Check Palindrome**	Strings	Very High	Easy	NQT

#	Question	Topic	Freq	Diff	Round
15	**Check Anagram**	Strings	Very High	Easy	NQT/Digital
16	**First Non-Repeating Character**	Strings	Very High	Easy	NQT/Digital
17	**Valid Parentheses**	Strings	Very High	Easy	NQT/Digital
18	**Longest Substring Without Repeating Characters**	Strings	Very High	Medium	Digital/Prime
19	**Two Sum using HashMap**	Hashing	Very High	Easy	NQT/Digital
20	**Reverse a Linked List**	Linked List	Very High	Easy	NQT/Digital
21	**Detect Loop in Linked List**	Linked List	Very High	Easy	NQT/Digital
22	**Find Middle of Linked List**	Linked List	Very High	Easy	NQT
23	**Merge Two Sorted Linked Lists**	Linked List	Very High	Easy	NQT/Digital
24	**Implement Stack using Array**	Stack	Very High	Easy	NQT
25	**Valid Parentheses Check**	Stack	Very High	Easy	NQT/Digital
26	**Next Greater Element**	Stack	Very High	Medium	NQT/Digital
27	**Implement Queue using Array**	Queue	Very High	Easy	NQT
28	**BFS of Graph using Queue**	Queue	Very High	Medium	NQT/Digital
29	**Level Order Traversal of Tree**	Queue	Very High	Easy	NQT/Digital
30	**Inorder Traversal (recursive & iterative)**	Trees	Very High	Easy	NQT/Digital
31	**Preorder Traversal**	Trees	Very High	Easy	NQT/Digital
32	**Postorder Traversal**	Trees	Very High	Easy	NQT/Digital
33	**Level Order Traversal**	Trees	Very High	Easy	NQT/Digital
34	**Height / Depth of Binary Tree**	Trees	Very High	Easy	NQT/Digital
35	**BST – Insert, Delete, Search**	Trees	Very High	Easy	NQT/Digital
36	**BFS of Graph**	Graphs	Very High	Easy	NQT/Digital

#	Question	Topic	Freq	Diff	Round
37	**DFS of Graph**	Graphs	Very High	Easy	NQT/Digital
38	**Detect Cycle in Undirected Graph (BFS/DFS)**	Graphs	Very High	Medium	Digital/Prime
39	**Detect Cycle in Directed Graph (DFS)**	Graphs	Very High	Medium	Digital/Prime
40	**Number of Islands (BFS/DFS on matrix)**	Graphs	Very High	Medium	NQT/Digital
41	**Fibonacci (recursive, memoized, iterative)**	Recursion	Very High	Easy	NQT
42	**Factorial**	Recursion	Very High	Easy	NQT
43	**0/1 Knapsack**	Dynamic Programming	Very High	Medium	Digital/Prime
44	**Longest Common Subsequence (LCS)**	Dynamic Programming	Very High	Medium	Digital/Prime
45	**Longest Increasing Subsequence (LIS)**	Dynamic Programming	Very High	Medium	Digital/Prime
46	**Coin Change – Minimum Coins**	Dynamic Programming	Very High	Medium	NQT/Digital/Prime
47	**Check Prime Number**	Number Theory	Very High	Easy	NQT
48	**GCD and LCM**	Number Theory	Very High	Easy	NQT
49	**Check Armstrong Number**	Number Theory	Very High	Easy	NQT
50	**Fibonacci Series (many methods)**	Number Theory	Very High	Easy	NQT
51	**Check Palindrome Number**	Number Theory	Very High	Easy	NQT
52	**Leaders in an Array**	Arrays	High	Easy	NQT/Digital
53	**Move All Zeroes to End**	Arrays	High	Easy	NQT
54	**Find Duplicates in Array**	Arrays	High	Easy	NQT/Digital
55	**Rotate Array by K positions**	Arrays	High	Easy	NQT
56	**Find Triplet with Given Sum (3-Sum)**	Arrays	High	Medium	Digital/Prime
57	**Merge Two Sorted Arrays**	Arrays	High	Easy	NQT/Digital
58	**Maximum Product Subarray**	Arrays	High	Medium	Digital/Prime

#	Question	Topic	Freq	Diff	Round
59	**Subarray with Given Sum**	Arrays	High	Medium	NQT/Digital
60	**Find Minimum in Rotated Sorted Array**	Arrays	High	Medium	Digital/Prime
61	**Search in Rotated Sorted Array**	Arrays	High	Medium	Digital/Prime
62	**Merge Intervals**	Arrays	High	Medium	Digital/Prime
63	**Trapping Rain Water**	Arrays	High	Hard	Digital/Prime
64	**Majority Element (Boyer-Moore)**	Arrays	High	Easy	NQT/Digital
65	**Find the Element that Appears Once (others appear twice)**	Arrays	High	Easy	NQT
66	**Find Kth Smallest / Largest Element**	Arrays	High	Medium	Digital/Prime
67	**Minimum Platforms for Train Schedule**	Arrays	High	Medium	Digital
68	**Maximum Sum of Non-Adjacent Elements**	Arrays	High	Medium	Digital
69	**Minimum Number of Jumps to Reach End**	Arrays	High	Medium	Digital/Prime
70	**Heap Sort Implementation**	Sorting	High	Medium	Digital
71	**Kth Largest Element using Quick Select**	Sorting	High	Medium	Digital/Prime
72	**Binary Search on Answer – Minimum Max Subarray**	Binary Search	High	Hard	Prime
73	**Floor and Ceiling in Sorted Array**	Binary Search	High	Easy	NQT/Digital
74	**Square Root using Binary Search**	Binary Search	High	Easy	NQT/Digital
75	**Peak Element in Array**	Binary Search	High	Medium	Digital
76	**Allocate Minimum Number of Pages**	Binary Search	High	Hard	Digital/Prime
77	**Aggressive Cows (Maximum Minimum Distance)**	Binary Search	High	Hard	Digital/Prime
78	**Count Vowels and Consonants**	Strings	High	Easy	NQT
79	**Remove Duplicate Characters**	Strings	High	Easy	NQT/Digital
80	**Longest Palindromic Substring**	Strings	High	Medium	Digital/Prime

#	Question	Topic	Freq	Diff	Round
81	**Longest Common Prefix**	Strings	High	Easy	NQT/Digital
82	**String Rotation Check**	Strings	High	Easy	NQT
83	**Roman to Integer**	Strings	High	Easy	NQT/Digital
84	**Group Anagrams Together**	Strings	High	Medium	Digital
85	**Compress String (Run Length Encoding)**	Strings	High	Easy	NQT
86	**Reverse Words in a String**	Strings	High	Easy	NQT/Digital
87	**Permutations of a String**	Strings	High	Medium	Digital
88	**Word Break Problem**	Strings	High	Medium	Digital/Prime
89	**Find All Pairs with Zero Sum**	Hashing	High	Easy	NQT/Digital
90	**Subarray with Given Sum (negative numbers)**	Hashing	High	Medium	Digital
91	**Find Longest Consecutive Sequence**	Hashing	High	Medium	Digital/Prime
92	**First Repeating Element**	Hashing	High	Easy	NQT
93	**Frequency of Each Element**	Hashing	High	Easy	NQT
94	**Find Start of Loop**	Linked List	High	Medium	Digital
95	**Remove Nth Node from End**	Linked List	High	Easy	NQT/Digital
96	**Check Palindrome Linked List**	Linked List	High	Medium	NQT/Digital
97	**Delete Node Without Head Pointer**	Linked List	High	Easy	Digital
98	**Add Two Numbers as Linked Lists**	Linked List	High	Medium	NQT/Digital
99	**Intersection Point of Two Linked Lists**	Linked List	High	Medium	NQT/Digital
100	**Reverse Linked List in Groups of K**	Linked List	High	Medium	Digital/Prime

MASTER DSA CHECKLIST — Tick Off As You Complete

Use this checklist during your daily practice. Mark each item once you can solve it from memory.

■ **[Arrays]** Find Second Largest Element	■ **[Arrays]** Find Missing Number (1 to N)
■ **[Arrays]** Kadane's Algorithm – Maximum Subarray Sum	■ **[Arrays]** Find Pair with Given Sum (Two Sum)
■ **[Arrays]** Best Time to Buy and Sell Stock	■ **[Arrays]** Sort Array of 0s 1s 2s (Dutch National Flag)
■ **[Sorting]** Bubble Sort Implementation	■ **[Sorting]** Selection Sort Implementation
■ **[Sorting]** Insertion Sort Implementation	■ **[Sorting]** Merge Sort Implementation
■ **[Sorting]** Quick Sort Implementation	■ **[Binary Search]** Binary Search – Standard
■ **[Strings]** Reverse a String	■ **[Strings]** Check Palindrome
■ **[Strings]** Check Anagram	■ **[Strings]** First Non-Repeating Character
■ **[Strings]** Valid Parentheses	■ **[Strings]** Longest Substring Without Repeating Characters
■ **[Hashing]** Two Sum using HashMap	■ **[Linked List]** Reverse a Linked List
■ **[Linked List]** Detect Loop in Linked List	■ **[Linked List]** Find Middle of Linked List
■ **[Linked List]** Merge Two Sorted Linked Lists	■ **[Stack]** Implement Stack using Array
■ **[Stack]** Valid Parentheses Check	■ **[Stack]** Next Greater Element
■ **[Queue]** Implement Queue using Array	■ **[Queue]** BFS of Graph using Queue
■ **[Queue]** Level Order Traversal of Tree	■ **[Trees]** Inorder Traversal (recursive & iterative)
■ **[Trees]** Preorder Traversal	■ **[Trees]** Postorder Traversal
■ **[Trees]** Level Order Traversal	■ **[Trees]** Height / Depth of Binary Tree
■ **[Trees]** BST – Insert, Delete, Search	■ **[Graphs]** BFS of Graph
■ **[Graphs]** DFS of Graph	■ **[Graphs]** Detect Cycle in Undirected Graph (BFS/DFS)
■ **[Graphs]** Detect Cycle in Directed Graph (DFS)	■ **[Graphs]** Number of Islands (BFS/DFS on matrix)
■ **[Recursion]** Fibonacci (recursive, memoized, iterative)	■ **[Recursion]** Factorial
■ **[Dynamic Programming]** 0/1 Knapsack	■ **[Dynamic Programming]** Longest Common Subsequence (LCS)
■ **[Dynamic Programming]** Longest Increasing Subsequence (LIS)	■ **[Dynamic Programming]** Coin Change – Minimum Coins
■ **[Number Theory]** Check Prime Number	■ **[Number Theory]** GCD and LCM
■ **[Number Theory]** Check Armstrong Number	■ **[Number Theory]** Fibonacci Series (many methods)
■ **[Number Theory]** Check Palindrome Number	■ **[Arrays]** Leaders in an Array
■ **[Arrays]** Move All Zeroes to End	■ **[Arrays]** Find Duplicates in Array
■ **[Arrays]** Rotate Array by K positions	■ **[Arrays]** Find Triplet with Given Sum (3-Sum)
■ **[Arrays]** Merge Two Sorted Arrays	■ **[Arrays]** Maximum Product Subarray
■ **[Arrays]** Subarray with Given Sum	■ **[Arrays]** Find Minimum in Rotated Sorted Array
■ **[Arrays]** Search in Rotated Sorted Array	■ **[Arrays]** Merge Intervals

■ **[Arrays]** Trapping Rain Water	■ **[Arrays]** Majority Element (Boyer-Moore)
■ **[Arrays]** Find the Element that Appears Once (others appear twice)	■ **[Arrays]** Find Kth Smallest / Largest Element
■ **[Arrays]** Minimum Platforms for Train Schedule	■ **[Arrays]** Maximum Sum of Non-Adjacent Elements
■ **[Arrays]** Minimum Number of Jumps to Reach End	■ **[Sorting]** Heap Sort Implementation
■ **[Sorting]** Kth Largest Element using Quick Select	■ **[Binary Search]** Binary Search on Answer – Minimum Max Subarray
■ **[Binary Search]** Floor and Ceiling in Sorted Array	■ **[Binary Search]** Square Root using Binary Search
■ **[Binary Search]** Peak Element in Array	■ **[Binary Search]** Allocate Minimum Number of Pages
■ **[Binary Search]** Aggressive Cows (Maximum Minimum Distance)	■ **[Strings]** Count Vowels and Consonants
■ **[Strings]** Remove Duplicate Characters	■ **[Strings]** Longest Palindromic Substring
■ **[Strings]** Longest Common Prefix	■ **[Strings]** String Rotation Check
■ **[Strings]** Roman to Integer	■ **[Strings]** Group Anagrams Together
■ **[Strings]** Compress String (Run Length Encoding)	■ **[Strings]** Reverse Words in a String
■ **[Strings]** Permutations of a String	■ **[Strings]** Word Break Problem
■ **[Hashing]** Find All Pairs with Zero Sum	■ **[Hashing]** Subarray with Given Sum (negative numbers)
■ **[Hashing]** Find Longest Consecutive Sequence	■ **[Hashing]** First Repeating Element
■ **[Hashing]** Frequency of Each Element	■ **[Linked List]** Find Start of Loop
■ **[Linked List]** Remove Nth Node from End	■ **[Linked List]** Check Palindrome Linked List
■ **[Linked List]** Delete Node Without Head Pointer	■ **[Linked List]** Add Two Numbers as Linked Lists
■ **[Linked List]** Intersection Point of Two Linked Lists	■ **[Linked List]** Reverse Linked List in Groups of K
■ **[Linked List]** Sort Linked List (Merge Sort)	■ **[Linked List]** LRU Cache Implementation
■ **[Stack]** Previous Smaller Element	■ **[Stack]** Stock Span Problem
■ **[Stack]** Largest Rectangle in Histogram	■ **[Stack]** Maximum Area Rectangle in Binary Matrix
■ **[Stack]** Celebrity Problem	■ **[Stack]** Sort a Stack (recursion)
■ **[Stack]** Evaluate Postfix Expression	■ **[Stack]** Infix to Postfix Conversion
■ **[Stack]** Min Stack (O(1) getMin)	■ **[Queue]** Implement Circular Queue
■ **[Queue]** Sliding Window Maximum using Deque	■ **[Queue]** First Non-Repeating Character in Stream
■ **[Queue]** Rotten Oranges – Multi-source BFS	■ **[Trees]** Diameter of Binary Tree
■ **[Trees]** Check if Binary Tree is Balanced	■ **[Trees]** Mirror / Invert Binary Tree
■ **[Trees]** Check if Two Trees are Identical	■ **[Trees]** Lowest Common Ancestor (BST)
■ **[Trees]** Lowest Common Ancestor (Binary Tree)	■ **[Trees]** Path from Root to Node
■ **[Trees]** Zigzag Level Order Traversal	■ **[Trees]** Vertical Order Traversal
■ **[Trees]** Top View of Binary Tree	■ **[Trees]** Bottom View of Binary Tree
■ **[Trees]** Left View / Right View of Binary Tree	■ **[Trees]** BST – Check if Valid BST
■ **[Trees]** BST – Kth Smallest Element	■ **[Trees]** Max Path Sum in Binary Tree

■ **[Trees]** Construct Tree from Inorder and Preorder	■ **[Trees]** Trie – Insert and Search
■ **[Graphs]** Topological Sort (DFS)	■ **[Graphs]** Topological Sort (Kahn's BFS)
■ **[Graphs]** Shortest Path – Dijkstra	■ **[Graphs]** Minimum Spanning Tree – Prim's
■ **[Graphs]** Minimum Spanning Tree – Kruskal's	■ **[Graphs]** Flood Fill Algorithm
■ **[Graphs]** Word Ladder – BFS shortest transform	■ **[Graphs]** Course Schedule I & II
■ **[Graphs]** Number of Connected Components	■ **[Recursion]** Tower of Hanoi
■ **[Recursion]** Generate All Subsets / Power Set	■ **[Recursion]** Generate All Permutations
■ **[Recursion]** N-Queens Problem	■ **[Recursion]** Rat in a Maze
■ **[Recursion]** Print All Paths from Source to Destination	■ **[Recursion]** Word Search in Matrix
■ **[Recursion]** Combination Sum I (target sum, reuse)	■ **[Recursion]** Subset Sum Problem
■ **[Recursion]** Generate Parentheses	■ **[Recursion]** Partition Equal Subset Sum
■ **[Recursion]** Flood Fill Recursive	■ **[Dynamic Programming]** Unbounded Knapsack
■ **[Dynamic Programming]** Edit Distance (Levenshtein)	■ **[Dynamic Programming]** Coin Change – Count Ways
■ **[Dynamic Programming]** Matrix Chain Multiplication	■ **[Dynamic Programming]** Word Break Problem
■ **[Dynamic Programming]** Rod Cutting Problem	■ **[Dynamic Programming]** Maximum Sum Increasing Subsequence
■ **[Dynamic Programming]** Print LCS	■ **[Dynamic Programming]** Minimum Path Sum in Grid
■ **[Dynamic Programming]** Unique Paths in Grid	■ **[Dynamic Programming]** Maximum Square of 1s in Matrix
■ **[Dynamic Programming]** House Robber I	■ **[Dynamic Programming]** Jump Game I (can reach end?)
■ **[Dynamic Programming]** Jump Game II (minimum jumps)	■ **[Bit Manipulation]** Count Set Bits in Integer
■ **[Bit Manipulation]** Check if Power of 2	■ **[Bit Manipulation]** Find XOR of All Numbers from 1 to N
■ **[Bit Manipulation]** Swap Two Numbers without Temp	■ **[Bit Manipulation]** Find the Two Non-Repeating Elements
■ **[Matrix]** Rotate Matrix 90 Degrees	■ **[Matrix]** Spiral Matrix Traversal
■ **[Matrix]** Search in Row-Column Sorted Matrix	■ **[Matrix]** Set Matrix Zeroes
■ **[Matrix]** Transpose of Matrix	■ **[Matrix]** Matrix Multiplication
■ **[Matrix]** Count Islands in Binary Matrix	■ **[Matrix]** Minimum Cost Path in Matrix
■ **[Number Theory]** Sieve of Eratosthenes	■ **[Number Theory]** Check Perfect Number
■ **[Number Theory]** Check Strong Number	■ **[Number Theory]** Power Function – Fast Exponentiation
■ **[Number Theory]** Modular Exponentiation	■ **[Number Theory]** Digit Sum / Digital Root
■ **[Number Theory]** Count Digits in Number	■ **[Number Theory]** Reverse Digits of Number
■ **[Number Theory]** Find All Divisors of N	■ **[Number Theory]** Pascal's Triangle
■ **[Arrays]** Find Intersection of Two Arrays	■ **[Arrays]** Find Union of Two Arrays
■ **[Arrays]** Equilibrium Index of Array	■ **[Arrays]** Count Inversions in Array

■ **[Arrays]** Best Time to Buy and Sell Stock II (multiple transactions)	■ **[Arrays]** Rearrange Array in Alternating +/- Sign
■ **[Arrays]** Find All Subarrays with Zero Sum	■ **[Arrays]** Longest Subarray with Equal 0s and 1s
■ **[Arrays]** Maximum Circular Subarray Sum	■ **[Arrays]** Count pairs with given difference K
■ **[Arrays]** Find the Element that Appears Once (others appear thrice)	■ **[Arrays]** Product of Array Except Self
■ **[Arrays]** Chocolate Distribution Problem	■ **[Arrays]** Next Permutation
■ **[Arrays]** Find First and Last Position of Element in Sorted Array	■ **[Arrays]** Count Occurrences of Element in Sorted Array
■ **[Arrays]** Wave Array	■ **[Arrays]** Segregate Even and Odd Numbers
■ **[Arrays]** Find the Median of Two Sorted Arrays	■ **[Sorting]** Counting Sort
■ **[Sorting]** Radix Sort	■ **[Sorting]** Sort Characters by Frequency
■ **[Sorting]** Sort Array by Parity	■ **[Sorting]** Custom Comparator Sorting – Sort by second element of pair
■ **[Binary Search]** Find Rotation Count in Rotated Sorted Array	■ **[Binary Search]** Median in Row-wise Sorted Matrix
■ **[Binary Search]** Kth Smallest Element in Sorted Matrix	■ **[Strings]** Count and Say Sequence
■ **[Strings]** Integer to Roman	■ **[Strings]** Longest Valid Parentheses
■ **[Strings]** Minimum Window Substring	■ **[Strings]** KMP String Search
■ **[Strings]** Isomorphic Strings	■ **[Strings]** Wildcard Pattern Matching
■ **[Strings]** Decode Ways	■ **[Strings]** Count Substrings with K Distinct Characters
■ **[Strings]** Print All Subsequences of a String	■ **[Hashing]** Count Distinct Elements in Every Window
■ **[Hashing]** 4-Sum Problem	■ **[Hashing]** Check if Array is Subset of Another
■ **[Linked List]** Merge K Sorted Linked Lists	■ **[Linked List]** Flatten a Multilevel Linked List
■ **[Linked List]** Clone Linked List with Random Pointer	■ **[Linked List]** Segregate Even and Odd Nodes
■ **[Linked List]** Rotate Linked List by K	■ **[Linked List]** Delete All Occurrences of a Key
■ **[Stack]** Implement Stack using Queue	■ **[Stack]** Implement Queue using Stack
■ **[Stack]** Reverse a Stack (recursion)	■ **[Stack]** Infix to Prefix Conversion
■ **[Stack]** Decode String (nested brackets)	■ **[Queue]** Interleave First and Second Half of Queue
■ **[Trees]** BST – Convert to Sorted DLL	■ **[Trees]** Serialize and Deserialize Binary Tree
■ **[Trees]** Boundary Traversal of Binary Tree	■ **[Trees]** Count Nodes in Complete Binary Tree
■ **[Trees]** Flatten Binary Tree to Linked List	■ **[Trees]** Construct Tree from Inorder and Postorder
■ **[Trees]** Segment Tree – Build and Query	■ **[Trees]** Trie – Longest Prefix Matching
■ **[Trees]** Check if Tree is Symmetric	■ **[Graphs]** Shortest Path – Bellman-Ford
■ **[Graphs]** Shortest Path – Floyd-Warshall (All Pairs)	■ **[Graphs]** Bipartite Check
■ **[Graphs]** Strongly Connected Components – Kosaraju	■ **[Graphs]** Clone a Graph
■ **[Graphs]** Alien Dictionary (Topological Sort)	■ **[Graphs]** Grid Shortest Path (0-1 BFS)
■ **[Graphs]** Snakes and Ladders – Shortest Path	■ **[Graphs]** Is Graph a Tree (no cycle + connected)

■ **[Recursion]** Sudoku Solver	■ **[Recursion]** Combination Sum II (no reuse)
■ **[Recursion]** Print K-th Permutation	■ **[Dynamic Programming]** Egg Drop Problem
■ **[Dynamic Programming]** Palindrome Partitioning – Minimum Cuts	■ **[Dynamic Programming]** Palindrome Partitioning – All Partitions
■ **[Dynamic Programming]** Longest Bitonic Subsequence	■ **[Dynamic Programming]** Shortest Common Supersequence
■ **[Dynamic Programming]** Distinct Subsequences	■ **[Dynamic Programming]** Wildcard Matching DP
■ **[Dynamic Programming]** Interleaving Strings	■ **[Dynamic Programming]** Unique Paths with Obstacles
■ **[Dynamic Programming]** Largest Rectangle in Histogram DP	■ **[Dynamic Programming]** House Robber II (circular)
■ **[Dynamic Programming]** Palindromic Substrings Count	■ **[Dynamic Programming]** Longest Palindromic Subsequence
■ **[Dynamic Programming]** Decode Ways	■ **[Bit Manipulation]** Reverse Bits of Integer
■ **[Bit Manipulation]** Find Position of Only Set Bit	■ **[Bit Manipulation]** Power Set using Bits
■ **[Bit Manipulation]** Hamming Distance	■ **[Bit Manipulation]** Add Two Numbers without + Operator
■ **[Matrix]** Scalar Multiplication of Matrix	■ **[Matrix]** Diagonal Sum of Matrix
■ **[Matrix]** Print Matrix in Zigzag	■ **[Matrix]** Find Maximum in each Row
■ **[Number Theory]** Catalan Number	■ **[Number Theory]** Ugly Numbers
■ **[Arrays]** Sliding Window Maximum	■ **[Arrays]** Array Rotation – Juggling Algorithm
■ **[Strings]** Rabin-Karp String Search	■ **[Number Theory]** Check Automorphic Number
■ **[Number Theory]** Check Neon Number	■ **[Arrays]** Maximum Length Bitonic Subarray
■ **[Arrays]** Count Total Set Bits in All Numbers 1 to N	■ **[Arrays]** Longest Mountain Subarray
■ **[Sorting]** Shell Sort	■ **[Sorting]** Pancake Sorting
■ **[Binary Search]** Find Position of Element in Infinite Sorted Array	■ **[Binary Search]** Search in Nearly Sorted Array
■ **[Strings]** Z Algorithm	■ **[Strings]** Regular Expression Matching
■ **[Hashing]** Group Integers with Same Digit Sum	■ **[Trees]** Fenwick Tree / BIT
■ **[Graphs]** Strongly Connected Components – Tarjan	■ **[Graphs]** Bridge and Articulation Point
■ **[Dynamic Programming]** Burst Balloons	■ **[Dynamic Programming]** Partition Array for Maximum Sum
■ **[Dynamic Programming]** Stock Buy Sell with Cooldown	■ **[Dynamic Programming]** Stock Buy Sell with Transaction Fee
■ **[Bit Manipulation]** Multiply by Power of 2	■ **[Bit Manipulation]** Divide by Power of 2
■ **[Number Theory]** Extended Euclidean Algorithm	

SECTION 2 SQL — Structured Query Language

Aggregate (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find the highest salary from Employee table	Aggregate	Very High	NQT/Digital	Easy
2	COUNT, SUM, AVG, MIN, MAX	Aggregate	Very High	NQT	Easy

Subquery/Window (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find the second highest salary	Subquery/Window	Very High	NQT/Digital/Prime	Medium
2	Find the Nth highest salary	Subquery/Window	Very High	Digital/Prime	Medium

GROUP BY/HAVING (3 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find duplicate rows in a table	GROUP BY/HAVING	Very High	NQT/Digital	Easy
2	Find departments with more than 5 employees	GROUP BY/HAVING	High	NQT/Digital	Easy
3	Write a query using GROUP BY and HAVING	GROUP BY/HAVING	Very High	NQT	Easy

DELETE/CTE (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Delete duplicate rows keeping one	DELETE/CTE	High	Digital/Prime	Medium

GROUP BY (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Count number of employees in each department	GROUP BY	Very High	NQT	Easy

Subquery (6 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees earning above their department average	Subquery	High	Digital/Prime	Medium

#	SQL Question	Topic	Freq	Round	Difficulty
2	Write a subquery in WHERE clause	Subquery	High	NQT/Digital	Medium
3	Write a correlated subquery	Subquery	High	Digital/Prime	Medium
4	EXISTS vs IN – difference with examples	Subquery	High	Digital	Medium
5	Find 3rd highest salary without using LIMIT or TOP	Subquery	High	Digital/Prime	Medium
6	Find median salary	Subquery	Medium	Prime	Hard

Date Functions (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees who joined in the last 6 months	Date Functions	High	Digital	Easy
2	Date functions: NOW(), DATEDIFF(), DATE_ADD(), EXTRACT()	Date Functions	High	Digital	Easy

Self Join/NULL (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees with no manager (top-level)	Self Join/NULL	High	Digital	Easy

Self Join (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees who are also managers	Self Join	High	Digital	Medium

Outer Join (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	List all employees and their managers (including those without managers)	Outer Join	High	Digital	Medium

GROUP BY/ORDER (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find the department with the highest total salary	GROUP BY/ORDER	High	Digital/Prime	Medium

Self Join / GROUP BY (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees who share the same salary	Self Join / GROUP BY	High	Digital	Medium

JOIN (7 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	INNER JOIN between Employee and Department	JOIN	Very High	NQT	Easy
2	LEFT JOIN – Show all employees even without department	JOIN	Very High	NQT/Digital	Easy
3	RIGHT JOIN usage	JOIN	High	NQT	Easy
4	FULL OUTER JOIN usage	JOIN	Medium	Digital	Easy
5	CROSS JOIN usage	JOIN	Medium	Digital	Easy
6	SELF JOIN – Employee-Manager relationship	JOIN	High	Digital	Medium
7	Explain NATURAL JOIN	JOIN	Medium	Digital	Easy

Clauses (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Difference between WHERE and HAVING (write queries for each)	Clauses	Very High	NQT/Digital	Easy

ORDER BY (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	ORDER BY multiple columns	ORDER BY	High	NQT	Easy

DISTINCT (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	DISTINCT keyword – find unique departments	DISTINCT	High	NQT	Easy

Pagination (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	LIMIT / TOP / ROWNUM – fetch top 5 records	Pagination	High	NQT/Digital	Easy

Operators (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Use of BETWEEN operator	Operators	High	NQT	Easy
2	Use of IN and NOT IN	Operators	High	NQT	Easy

String Ops (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Use of LIKE for pattern matching	String Ops	High	NQT	Easy

NULL Handling (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Use of IS NULL and IS NOT NULL	NULL Handling	High	NQT	Easy

Set Operations (4 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Write a query using UNION	Set Operations	High	NQT/Digital	Easy
2	UNION vs UNION ALL difference	Set Operations	High	NQT/Digital	Easy
3	INTERSECT and EXCEPT / MINUS operators	Set Operations	Medium	Digital	Easy
4	Find employees who appear in both 2021 and 2022 tables	Set Operations	Medium	Digital	Medium

Window Functions (6 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	ROW_NUMBER() window function	Window Functions	High	Digital/Prime	Medium
2	RANK() and DENSE_RANK() difference	Window Functions	Very High	Digital/Prime	Medium
3	LEAD() and LAG() functions	Window Functions	High	Digital/Prime	Medium
4	NTILE() function	Window Functions	Low	Prime	Medium
5	PARTITION BY with window functions	Window Functions	High	Digital/Prime	Medium
6	Running total using SUM() OVER	Window Functions	High	Digital/Prime	Medium

Views (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Create a VIEW	Views	High	Digital	Easy
2	Updatable View – can we update through a view?	Views	Medium	Digital	Medium

Stored Proc (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Create a stored procedure with parameters	Stored Proc	High	Digital/Prime	Medium

Triggers (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Create a TRIGGER (AFTER INSERT)	Triggers	High	Digital/Prime	Medium
2	BEFORE vs AFTER trigger	Triggers	Medium	Digital	Easy

Indexing (3 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Create an INDEX and explain its types	Indexing	High	Digital/Prime	Medium
2	Clustered vs Non-Clustered Index	Indexing	Very High	Digital/Prime	Medium
3	Drop an index	Indexing	Medium	Digital	Easy

Constraints (6 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Primary Key constraint	Constraints	Very High	NQT	Easy
2	Foreign Key constraint and CASCADE	Constraints	Very High	NQT/Digital	Easy
3	UNIQUE constraint	Constraints	High	NQT	Easy
4	NOT NULL constraint	Constraints	High	NQT	Easy
5	CHECK constraint	Constraints	Medium	Digital	Easy
6	DEFAULT constraint	Constraints	Medium	NQT	Easy

DDL (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	ALTER TABLE – add / drop / modify column	DDL	High	NQT/Digital	Easy
2	TRUNCATE vs DELETE vs DROP	DDL	Very High	NQT/Digital	Easy

Transactions (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Transactions – BEGIN, COMMIT, ROLLBACK	Transactions	High	Digital/Prime	Medium
2	SAVEPOINT usage in transactions	Transactions	Medium	Digital	Medium

String/LIKE (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees whose names start with 'A'	String/LIKE	High	NQT	Easy

Date (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find employees hired in year 2022	Date	High	NQT/Digital	Easy

Type Conversion (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Convert string to date (CAST / CONVERT)	Type Conversion	Medium	Digital	Easy

Window (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find cumulative salary department-wise	Window	Medium	Prime	Medium

CTE (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	CTE (Common Table Expression) usage	CTE	High	Digital/Prime	Medium
2	Recursive CTE – hierarchy query	CTE	Medium	Prime	Hard

Advanced (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Pivot table in SQL	Advanced	Medium	Prime	Hard

NULL (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Handle NULL in aggregate: COALESCE, IFNULL, NVL	NULL	High	Digital	Easy

String Functions (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	String functions: UPPER, LOWER, SUBSTR, LENGTH, TRIM, REPLACE	String Functions	High	NQT/Digital	Easy

Control Flow (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	CASE WHEN statement	Control Flow	High	NQT/Digital	Easy
2	IIF and NULLIF	Control Flow	Medium	Digital	Easy

Window/JOIN (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Write a query to rank employees by salary within department	Window/JOIN	High	Digital/Prime	Medium

DML (2 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Insert multiple rows with single INSERT	DML	Medium	NQT	Easy

#	SQL Question	Topic	Freq	Round	Difficulty
2	Update salary of all employees in a department by 10 percent	DML	High	NQT	Easy

Data Types (1 questions)

#	SQL Question	Topic	Freq	Round	Difficulty
1	Difference between CHAR and VARCHAR	Data Types	Very High	NQT/Digital	Easy

TOP 25 — SQL — Structured Query Language

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find the highest salary from Employee table	Aggregate	Very High	NQT/Digital	Easy
2	Find the second highest salary	Subquery/Window	Very High	NQT/Digital/Prime	Medium
3	Find the Nth highest salary	Subquery/Window	Very High	Digital/Prime	Medium
4	Find duplicate rows in a table	GROUP BY/HAVING	Very High	NQT/Digital	Easy
5	Count number of employees in each department	GROUP BY	Very High	NQT	Easy
6	INNER JOIN between Employee and Department	JOIN	Very High	NQT	Easy
7	LEFT JOIN – Show all employees even without department	JOIN	Very High	NQT/Digital	Easy
8	Write a query using GROUP BY and HAVING	GROUP BY/HAVING	Very High	NQT	Easy
9	Difference between WHERE and HAVING (write queries for each)	Clauses	Very High	NQT/Digital	Easy
10	COUNT, SUM, AVG, MIN, MAX	Aggregate	Very High	NQT	Easy
11	RANK() and DENSE_RANK() difference	Window Functions	Very High	Digital/Prime	Medium
12	Clustered vs Non-Clustered Index	Indexing	Very High	Digital/Prime	Medium
13	Primary Key constraint	Constraints	Very High	NQT	Easy
14	Foreign Key constraint and CASCADE	Constraints	Very High	NQT/Digital	Easy
15	TRUNCATE vs DELETE vs DROP	DDL	Very High	NQT/Digital	Easy
16	Difference between CHAR and VARCHAR	Data Types	Very High	NQT/Digital	Easy
17	Delete duplicate rows keeping one	DELETE/CTE	High	Digital/Prime	Medium
18	Find employees earning above their department average	Subquery	High	Digital/Prime	Medium
19	Find departments with more than 5 employees	GROUP BY/HAVING	High	NQT/Digital	Easy
20	Find employees who joined in the last 6 months	Date Functions	High	Digital	Easy

#	SQL Question	Topic	Freq	Round	Difficulty
21	Find employees with no manager (top-level)	Self Join/NULL	High	Digital	Easy
22	Find employees who are also managers	Self Join	High	Digital	Medium
23	List all employees and their managers (including those without managers)	Outer Join	High	Digital	Medium
24	Find the department with the highest total salary	GROUP BY/ORDER	High	Digital/Prime	Medium
25	Find employees who share the same salary	Self Join / GROUP BY	High	Digital	Medium

TOP 50 — SQL — Structured Query Language

#	SQL Question	Topic	Freq	Round	Difficulty
1	Find the highest salary from Employee table	Aggregate	Very High	NQT/Digital	Easy
2	Find the second highest salary	Subquery/Window	Very High	NQT/Digital/Prime	Medium
3	Find the Nth highest salary	Subquery/Window	Very High	Digital/Prime	Medium
4	Find duplicate rows in a table	GROUP BY/HAVING	Very High	NQT/Digital	Easy
5	Count number of employees in each department	GROUP BY	Very High	NQT	Easy
6	INNER JOIN between Employee and Department	JOIN	Very High	NQT	Easy
7	LEFT JOIN – Show all employees even without department	JOIN	Very High	NQT/Digital	Easy
8	Write a query using GROUP BY and HAVING	GROUP BY/HAVING	Very High	NQT	Easy
9	Difference between WHERE and HAVING (write queries for each)	Clauses	Very High	NQT/Digital	Easy
10	COUNT, SUM, AVG, MIN, MAX	Aggregate	Very High	NQT	Easy
11	RANK() and DENSE_RANK() difference	Window Functions	Very High	Digital/Prime	Medium
12	Clustered vs Non-Clustered Index	Indexing	Very High	Digital/Prime	Medium
13	Primary Key constraint	Constraints	Very High	NQT	Easy
14	Foreign Key constraint and CASCADE	Constraints	Very High	NQT/Digital	Easy

#	SQL Question	Topic	Freq	Round	Difficulty
15	TRUNCATE vs DELETE vs DROP	DDL	Very High	NQT/Digital	Easy
16	Difference between CHAR and VARCHAR	Data Types	Very High	NQT/Digital	Easy
17	Delete duplicate rows keeping one	DELETE/CTE	High	Digital/Prime	Medium
18	Find employees earning above their department average	Subquery	High	Digital/Prime	Medium
19	Find departments with more than 5 employees	GROUP BY/HAVING	High	NQT/Digital	Easy
20	Find employees who joined in the last 6 months	Date Functions	High	Digital	Easy
21	Find employees with no manager (top-level)	Self Join/NULL	High	Digital	Easy
22	Find employees who are also managers	Self Join	High	Digital	Medium
23	List all employees and their managers (including those without managers)	Outer Join	High	Digital	Medium
24	Find the department with the highest total salary	GROUP BY/ORDER	High	Digital/Prime	Medium
25	Find employees who share the same salary	Self Join / GROUP BY	High	Digital	Medium
26	RIGHT JOIN usage	JOIN	High	NQT	Easy
27	SELF JOIN – Employee-Manager relationship	JOIN	High	Digital	Medium
28	ORDER BY multiple columns	ORDER BY	High	NQT	Easy
29	DISTINCT keyword – find unique departments	DISTINCT	High	NQT	Easy
30	LIMIT / TOP / ROWNUM – fetch top 5 records	Pagination	High	NQT/Digital	Easy
31	Use of BETWEEN operator	Operators	High	NQT	Easy
32	Use of IN and NOT IN	Operators	High	NQT	Easy
33	Use of LIKE for pattern matching	String Ops	High	NQT	Easy
34	Use of IS NULL and IS NOT NULL	NULL Handling	High	NQT	Easy
35	Write a subquery in WHERE clause	Subquery	High	NQT/Digital	Medium
36	Write a correlated subquery	Subquery	High	Digital/Prime	Medium

#	SQL Question	Topic	Freq	Round	Difficulty
37	EXISTS vs IN – difference with examples	Subquery	High	Digital	Medium
38	Write a query using UNION	Set Operations	High	NQT/Digital	Easy
39	UNION vs UNION ALL difference	Set Operations	High	NQT/Digital	Easy
40	ROW_NUMBER() window function	Window Functions	High	Digital/Prime	Medium
41	LEAD() and LAG() functions	Window Functions	High	Digital/Prime	Medium
42	PARTITION BY with window functions	Window Functions	High	Digital/Prime	Medium
43	Running total using SUM() OVER	Window Functions	High	Digital/Prime	Medium
44	Create a VIEW	Views	High	Digital	Easy
45	Create a stored procedure with parameters	Stored Proc	High	Digital/Prime	Medium
46	Create a TRIGGER (AFTER INSERT)	Triggers	High	Digital/Prime	Medium
47	Create an INDEX and explain its types	Indexing	High	Digital/Prime	Medium
48	UNIQUE constraint	Constraints	High	NQT	Easy
49	NOT NULL constraint	Constraints	High	NQT	Easy
50	ALTER TABLE – add / drop / modify column	DDL	High	NQT/Digital	Easy

MASTER SQL — STRUCTURED QUERY LANGUAGE CHECKLIST

■ **[Aggregate]** Find the highest salary from Employee table	■ **[Subquery/Window]** Find the second highest salary
■ **[Subquery/Window]** Find the Nth highest salary	■ **[GROUP BY/HAVING]** Find duplicate rows in a table
■ **[GROUP BY]** Count number of employees in each department	■ **[JOIN]** INNER JOIN between Employee and Department
■ **[JOIN]** LEFT JOIN – Show all employees even without department	■ **[GROUP BY/HAVING]** Write a query using GROUP BY and HAVING
■ **[Clauses]** Difference between WHERE and HAVING (write queries for each)	■ **[Aggregate]** COUNT, SUM, AVG, MIN, MAX
■ **[Window Functions]** RANK() and DENSE_RANK() difference	■ **[Indexing]** Clustered vs Non-Clustered Index
■ **[Constraints]** Primary Key constraint	■ **[Constraints]** Foreign Key constraint and CASCADE
■ **[DDL]** TRUNCATE vs DELETE vs DROP	■ **[Data Types]** Difference between CHAR and VARCHAR
■ **[DELETE/CTE]** Delete duplicate rows keeping one	■ **[Subquery]** Find employees earning above their department average
■ **[GROUP BY/HAVING]** Find departments with more than 5 employees	■ **[Date Functions]** Find employees who joined in the last 6 months
■ **[Self Join/NULL]** Find employees with no manager (top-level)	■ **[Self Join]** Find employees who are also managers
■ **[Outer Join]** List all employees and their managers (including those without managers)	■ **[GROUP BY/ORDER]** Find the department with the highest total salary
■ **[Self Join / GROUP BY]** Find employees who share the same salary	■ **[JOIN]** RIGHT JOIN usage
■ **[JOIN]** SELF JOIN – Employee-Manager relationship	■ **[ORDER BY]** ORDER BY multiple columns
■ **[DISTINCT]** DISTINCT keyword – find unique departments	■ **[Pagination]** LIMIT / TOP / ROWNUM – fetch top 5 records
■ **[Operators]** Use of BETWEEN operator	■ **[Operators]** Use of IN and NOT IN
■ **[String Ops]** Use of LIKE for pattern matching	■ **[NULL Handling]** Use of IS NULL and IS NOT NULL
■ **[Subquery]** Write a subquery in WHERE clause	■ **[Subquery]** Write a correlated subquery
■ **[Subquery]** EXISTS vs IN – difference with examples	■ **[Set Operations]** Write a query using UNION
■ **[Set Operations]** UNION vs UNION ALL difference	■ **[Window Functions]** ROW_NUMBER() window function
■ **[Window Functions]** LEAD() and LAG() functions	■ **[Window Functions]** PARTITION BY with window functions
■ **[Window Functions]** Running total using SUM() OVER	■ **[Views]** Create a VIEW
■ **[Stored Proc]** Create a stored procedure with parameters	■ **[Triggers]** Create a TRIGGER (AFTER INSERT)
■ **[Indexing]** Create an INDEX and explain its types	■ **[Constraints]** UNIQUE constraint

■ **[Constraints]** NOT NULL constraint	■ **[DDL]** ALTER TABLE – add / drop / modify column
■ **[Transactions]** Transactions – BEGIN, COMMIT, ROLLBACK	■ **[Subquery]** Find 3rd highest salary without using LIMIT or TOP
■ **[String/LIKE]** Find employees whose names start with 'A'	■ **[Date]** Find employees hired in year 2022
■ **[CTE]** CTE (Common Table Expression) usage	■ **[NULL]** Handle NULL in aggregate: COALESCE, IFNULL, NVL
■ **[String Functions]** String functions: UPPER, LOWER, SUBSTR, LENGTH, TRIM, REPLACE	■ **[Date Functions]** Date functions: NOW(), DATEDIFF(), DATE_ADD(), EXTRACT()
■ **[Control Flow]** CASE WHEN statement	■ **[Window/JOIN]** Write a query to rank employees by salary within department
■ **[DML]** Update salary of all employees in a department by 10 percent	■ **[JOIN]** FULL OUTER JOIN usage
■ **[JOIN]** CROSS JOIN usage	■ **[Set Operations]** INTERSECT and EXCEPT / MINUS operators
■ **[Views]** Updatable View – can we update through a view?	■ **[Triggers]** BEFORE vs AFTER trigger
■ **[Indexing]** Drop an index	■ **[Constraints]** CHECK constraint
■ **[Constraints]** DEFAULT constraint	■ **[Transactions]** SAVEPOINT usage in transactions
■ **[Type Conversion]** Convert string to date (CAST / CONVERT)	■ **[Window]** Find cumulative salary department-wise
■ **[Subquery]** Find median salary	■ **[CTE]** Recursive CTE – hierarchy query
■ **[Advanced]** Pivot table in SQL	■ **[Control Flow]** IIF and NULLIF
■ **[Set Operations]** Find employees who appear in both 2021 and 2022 tables	■ **[DML]** Insert multiple rows with single INSERT
■ **[JOIN]** Explain NATURAL JOIN	■ **[Window Functions]** NTILE() function

SECTION 3 DBMS — Database Management Systems

Fundamentals (4 questions)

#	Question	Topic	Freq	Round	Interview
1	What is DBMS? Advantages over file system?	Fundamentals	Very High	NQT/Digital	Interview
2	Difference between DBMS and RDBMS	Fundamentals	Very High	NQT/Digital	Interview
3	What is a Schema? Types of schemas?	Fundamentals	High	NQT/Digital	Interview
4	What is Data Independence?	Fundamentals	High	Digital	Interview

Architecture (1 questions)

#	Question	Topic	Freq	Round	Interview
1	Explain 3-tier / 3-level architecture of DBMS	Architecture	High	Digital	Interview

Relational Model (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Relation? Difference between table and relation?	Relational Model	High	NQT/Digital	Interview

Keys (8 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Primary Key?	Keys	Very High	NQT	Interview
2	What is a Candidate Key?	Keys	Very High	NQT/Digital	Interview
3	What is a Super Key?	Keys	High	Digital	Interview
4	What is a Foreign Key? Give example.	Keys	Very High	NQT/Digital	Interview
5	What is an Alternate Key?	Keys	High	NQT	Interview
6	What is a Composite Key?	Keys	High	NQT/Digital	Interview
7	What is a Surrogate Key?	Keys	Medium	Digital	Interview
8	Difference between Primary Key and Unique Key	Keys	Very High	NQT/Digital	Interview

Constraints (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Referential Integrity?	Constraints	High	Digital	Interview

#	Question	Topic	Freq	Round	Interview
2	What is a NULL value? Is NULL = NULL?	Constraints	High	NQT/Digital	Interview

Transactions (8 questions)

#	Question	Topic	Freq	Round	Interview
1	What are ACID properties? Explain each.	Transactions	Very High	Digital/Prime	Interview
2	What is Atomicity?	Transactions	High	Digital	Interview
3	What is Consistency in database?	Transactions	High	Digital	Interview
4	What is Isolation? Isolation levels?	Transactions	High	Digital/Prime	Interview
5	What is Durability?	Transactions	High	Digital	Interview
6	What is a Transaction?	Transactions	Very High	Digital/Prime	Interview
7	States of a Transaction (Active, Partially Committed, etc.)	Transactions	Medium	Digital	Interview
8	What is Commit and Rollback?	Transactions	High	NQT/Digital	Interview

Concurrency (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Deadlock in DBMS?	Concurrency	High	Digital/Prime	Interview
2	Deadlock prevention vs avoidance vs detection	Concurrency	High	Digital/Prime	Interview
3	What is a Lock? Types of locks.	Concurrency	High	Digital/Prime	Interview
4	Shared Lock vs Exclusive Lock	Concurrency	High	Digital/Prime	Interview
5	What is Two-Phase Locking (2PL)?	Concurrency	Medium	Prime	Interview
6	What is Optimistic Concurrency Control?	Concurrency	Low	Prime	Interview
7	What is Timestamp Ordering protocol?	Concurrency	Low	Prime	Interview

Normalization (8 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Normalization? Why is it needed?	Normalization	Very High	NQT/Digital/Prime	Interview
2	Explain 1NF with example	Normalization	Very High	NQT/Digital	Interview
3	Explain 2NF with example	Normalization	Very High	NQT/Digital	Interview
4	Explain 3NF with example	Normalization	Very High	Digital/Prime	Interview
5	Explain BCNF with example	Normalization	Very High	Digital/Prime	Interview
6	Explain 4NF	Normalization	Medium	Prime	Interview
7	Explain 5NF	Normalization	Low	Prime	Interview
8	What is Denormalization? When to use it?	Normalization	High	Digital/Prime	Interview

FD (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Functional Dependency?	FD	High	NQT/Digital	Interview
2	Trivial vs Non-trivial Functional Dependency	FD	Medium	Digital	Interview
3	What is Full Functional Dependency?	FD	Medium	Digital	Interview
4	What is Partial Dependency?	FD	High	NQT/Digital	Interview
5	What is Transitive Dependency?	FD	High	NQT/Digital	Interview
6	What is Closure of Attribute Set?	FD	Medium	Digital	Interview
7	What is a Multi-valued Dependency?	FD	Low	Prime	Interview

ER Model (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is an ER Diagram? Draw ER for a university.	ER Model	Very High	NQT/Digital	Interview
2	Entities vs Attributes vs Relationships in ER	ER Model	High	NQT/Digital	Interview
3	Weak Entity vs Strong Entity	ER Model	High	Digital	Interview
4	Cardinality in ER (1:1, 1:N, M:N)	ER Model	High	NQT/Digital	Interview
5	Participation constraints (total vs partial)	ER Model	Medium	Digital	Interview
6	Convert ER Diagram to Relational Schema	ER Model	High	Digital/Prime	Interview

Indexing (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Indexing? Types of indexes.	Indexing	Very High	Digital/Prime	Interview
2	Clustered vs Non-Clustered Index	Indexing	Very High	Digital/Prime	Interview
3	Dense Index vs Sparse Index	Indexing	Medium	Digital	Interview
4	What is B-Tree Index?	Indexing	High	Digital/Prime	Interview
5	What is B+ Tree? Advantages over B-Tree?	Indexing	High	Digital/Prime	Interview
6	What is a Hash Index?	Indexing	Medium	Digital	Interview

Views (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a View? Advantages?	Views	High	Digital	Interview
2	Difference between View and Table	Views	High	Digital	Interview

Procedures (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Stored Procedure?	Procedures	Medium	Digital	Interview
2	Stored Procedure vs Function	Procedures	Medium	Digital	Interview

Triggers (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Trigger?	Triggers	Medium	Digital	Interview

Cursors (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Cursor?	Cursors	Medium	Digital	Interview

Relational Algebra (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Relational Algebra? Operators.	Relational Algebra	High	Digital/Prime	Interview
2	Selection vs Projection in Relational Algebra	Relational Algebra	High	Digital	Interview
3	What is a Join in Relational Algebra?	Relational Algebra	High	Digital	Interview

Query Processing (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Query Optimization?	Query Processing	Medium	Prime	Interview
2	What is a Query Execution Plan?	Query Processing	Medium	Prime	Interview

Advanced (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Data Warehouse?	Advanced	Medium	Digital/Prime	Interview
2	OLTP vs OLAP	Advanced	High	Digital/Prime	Interview

Distributed DB (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Sharding?	Distributed DB	Medium	Prime	Interview
2	What is Replication in databases?	Distributed DB	Medium	Prime	Interview
3	CAP Theorem	Distributed DB	Medium	Prime	Interview

NoSQL (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is NoSQL? Types of NoSQL databases?	NoSQL	Medium	Prime	Interview
2	SQL vs NoSQL – when to use which?	NoSQL	Medium	Prime	Interview

Recovery (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Recovery in DBMS? Techniques?	Recovery	High	Digital/Prime	Interview
2	Write-Ahead Logging (WAL)	Recovery	Medium	Prime	Interview
3	What is a Checkpoint in DBMS?	Recovery	Medium	Digital	Interview

SQL-DBMS (4 questions)

#	Question	Topic	Freq	Round	Interview
1	What is the difference between DROP, TRUNCATE, DELETE?	SQL-DBMS	Very High	NQT/Digital	Interview
2	Explain different types of JOINs	SQL-DBMS	Very High	NQT/Digital	Interview
3	What is a Self-Join?	SQL-DBMS	High	Digital	Interview
4	Difference between HAVING and WHERE	SQL-DBMS	Very High	NQT/Digital	Interview

TOP 25 — DBMS — Database Management Systems

#	Question	Topic	Freq	Round	Interview
1	What is DBMS? Advantages over file system?	Fundamentals	Very High	NQT/Digital	Interview
2	Difference between DBMS and RDBMS	Fundamentals	Very High	NQT/Digital	Interview
3	What is a Primary Key?	Keys	Very High	NQT	Interview
4	What is a Candidate Key?	Keys	Very High	NQT/Digital	Interview
5	What is a Foreign Key? Give example.	Keys	Very High	NQT/Digital	Interview
6	Difference between Primary Key and Unique Key	Keys	Very High	NQT/Digital	Interview
7	What are ACID properties? Explain each.	Transactions	Very High	Digital/Prime	Interview
8	What is a Transaction?	Transactions	Very High	Digital/Prime	Interview
9	What is Normalization? Why is it needed?	Normalization	Very High	NQT/Digital/Prime	Interview
10	Explain 1NF with example	Normalization	Very High	NQT/Digital	Interview
11	Explain 2NF with example	Normalization	Very High	NQT/Digital	Interview
12	Explain 3NF with example	Normalization	Very High	Digital/Prime	Interview
13	Explain BCNF with example	Normalization	Very High	Digital/Prime	Interview
14	What is an ER Diagram? Draw ER for a university.	ER Model	Very High	NQT/Digital	Interview
15	What is Indexing? Types of indexes.	Indexing	Very High	Digital/Prime	Interview
16	Clustered vs Non-Clustered Index	Indexing	Very High	Digital/Prime	Interview
17	What is the difference between DROP, TRUNCATE, DELETE?	SQL-DBMS	Very High	NQT/Digital	Interview
18	Explain different types of JOINs	SQL-DBMS	Very High	NQT/Digital	Interview
19	Difference between HAVING and WHERE	SQL-DBMS	Very High	NQT/Digital	Interview
20	What is a Schema? Types of schemas?	Fundamentals	High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
21	What is Data Independence?	Fundamentals	High	Digital	Interview
22	Explain 3-tier / 3-level architecture of DBMS	Architecture	High	Digital	Interview
23	What is a Relation? Difference between table and relation?	Relational Model	High	NQT/Digital	Interview
24	What is a Super Key?	Keys	High	Digital	Interview
25	What is an Alternate Key?	Keys	High	NQT	Interview

TOP 50 — DBMS — Database Management Systems

#	Question	Topic	Freq	Round	Interview
1	What is DBMS? Advantages over file system?	Fundamentals	Very High	NQT/Digital	Interview
2	Difference between DBMS and RDBMS	Fundamentals	Very High	NQT/Digital	Interview
3	What is a Primary Key?	Keys	Very High	NQT	Interview
4	What is a Candidate Key?	Keys	Very High	NQT/Digital	Interview
5	What is a Foreign Key? Give example.	Keys	Very High	NQT/Digital	Interview
6	Difference between Primary Key and Unique Key	Keys	Very High	NQT/Digital	Interview
7	What are ACID properties? Explain each.	Transactions	Very High	Digital/Prime	Interview
8	What is a Transaction?	Transactions	Very High	Digital/Prime	Interview
9	What is Normalization? Why is it needed?	Normalization	Very High	NQT/Digital/Prime	Interview
10	Explain 1NF with example	Normalization	Very High	NQT/Digital	Interview
11	Explain 2NF with example	Normalization	Very High	NQT/Digital	Interview
12	Explain 3NF with example	Normalization	Very High	Digital/Prime	Interview
13	Explain BCNF with example	Normalization	Very High	Digital/Prime	Interview
14	What is an ER Diagram? Draw ER for a university.	ER Model	Very High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
15	What is Indexing? Types of indexes.	Indexing	Very High	Digital/Prime	Interview
16	Clustered vs Non-Clustered Index	Indexing	Very High	Digital/Prime	Interview
17	What is the difference between DROP, TRUNCATE, DELETE?	SQL-DBMS	Very High	NQT/Digital	Interview
18	Explain different types of JOINs	SQL-DBMS	Very High	NQT/Digital	Interview
19	Difference between HAVING and WHERE	SQL-DBMS	Very High	NQT/Digital	Interview
20	What is a Schema? Types of schemas?	Fundamentals	High	NQT/Digital	Interview
21	What is Data Independence?	Fundamentals	High	Digital	Interview
22	Explain 3-tier / 3-level architecture of DBMS	Architecture	High	Digital	Interview
23	What is a Relation? Difference between table and relation?	Relational Model	High	NQT/Digital	Interview
24	What is a Super Key?	Keys	High	Digital	Interview
25	What is an Alternate Key?	Keys	High	NQT	Interview
26	What is a Composite Key?	Keys	High	NQT/Digital	Interview
27	What is Referential Integrity?	Constraints	High	Digital	Interview
28	What is a NULL value? Is NULL = NULL?	Constraints	High	NQT/Digital	Interview
29	What is Atomicity?	Transactions	High	Digital	Interview
30	What is Consistency in database?	Transactions	High	Digital	Interview
31	What is Isolation? Isolation levels?	Transactions	High	Digital/Prime	Interview
32	What is Durability?	Transactions	High	Digital	Interview
33	What is Commit and Rollback?	Transactions	High	NQT/Digital	Interview
34	What is a Deadlock in DBMS?	Concurrency	High	Digital/Prime	Interview
35	Deadlock prevention vs avoidance vs detection	Concurrency	High	Digital/Prime	Interview
36	What is a Lock? Types of locks.	Concurrency	High	Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
37	Shared Lock vs Exclusive Lock	Concurrency	High	Digital/Prime	Interview
38	What is Denormalization? When to use it?	Normalization	High	Digital/Prime	Interview
39	What is Functional Dependency?	FD	High	NQT/Digital	Interview
40	What is Partial Dependency?	FD	High	NQT/Digital	Interview
41	What is Transitive Dependency?	FD	High	NQT/Digital	Interview
42	Entities vs Attributes vs Relationships in ER	ER Model	High	NQT/Digital	Interview
43	Weak Entity vs Strong Entity	ER Model	High	Digital	Interview
44	Cardinality in ER (1:1, 1:N, M:N)	ER Model	High	NQT/Digital	Interview
45	Convert ER Diagram to Relational Schema	ER Model	High	Digital/Prime	Interview
46	What is B-Tree Index?	Indexing	High	Digital/Prime	Interview
47	What is B+ Tree? Advantages over B-Tree?	Indexing	High	Digital/Prime	Interview
48	What is a View? Advantages?	Views	High	Digital	Interview
49	Difference between View and Table	Views	High	Digital	Interview
50	What is Relational Algebra? Operators.	Relational Algebra	High	Digital/Prime	Interview

MASTER DBMS — DATABASE MANAGEMENT SYSTEMS CHECKLIST

- ■ **[Fundamentals]** What is DBMS? Advantages over file system?
- ■ **[Fundamentals]** Difference between DBMS and RDBMS
- ■ **[Keys]** What is a Primary Key?
- ■ **[Keys]** What is a Candidate Key?
- ■ **[Keys]** What is a Foreign Key? Give example.
- ■ **[Keys]** Difference between Primary Key and Unique Key
- ■ **[Transactions]** What are ACID properties? Explain each.
- ■ **[Transactions]** What is a Transaction?
- ■ **[Normalization]** What is Normalization? Why is it needed?
- ■ **[Normalization]** Explain 1NF with example
- ■ **[Normalization]** Explain 2NF with example
- ■ **[Normalization]** Explain 3NF with example
- ■ **[Normalization]** Explain BCNF with example
- ■ **[ER Model]** What is an ER Diagram? Draw ER for a university.
- ■ **[Indexing]** What is Indexing? Types of indexes.
- ■ **[Indexing]** Clustered vs Non-Clustered Index
- ■ **[SQL-DBMS]** What is the difference between DROP, TRUNCATE, DELETE?
- ■ **[SQL-DBMS]** Explain different types of JOINs
- ■ **[SQL-DBMS]** Difference between HAVING and WHERE
- ■ **[Fundamentals]** What is a Schema? Types of schemas?
- ■ **[Fundamentals]** What is Data Independence?
- ■ **[Architecture]** Explain 3-tier / 3-level architecture of DBMS
- ■ **[Relational Model]** What is a Relation? Difference between table and relation?
- ■ **[Keys]** What is a Super Key?
- ■ **[Keys]** What is an Alternate Key?
- ■ **[Keys]** What is a Composite Key?
- ■ **[Constraints]** What is Referential Integrity?
- ■ **[Constraints]** What is a NULL value? Is NULL = NULL?
- ■ **[Transactions]** What is Atomicity?
- ■ **[Transactions]** What is Consistency in database?
- ■ **[Transactions]** What is Isolation? Isolation levels?
- ■ **[Transactions]** What is Durability?
- ■ **[Transactions]** What is Commit and Rollback?
- ■ **[Concurrency]** What is a Deadlock in DBMS?
- ■ **[Concurrency]** Deadlock prevention vs avoidance vs detection
- ■ **[Concurrency]** What is a Lock? Types of locks.
- ■ **[Concurrency]** Shared Lock vs Exclusive Lock
- ■ **[Normalization]** What is Denormalization? When to use it?
- ■ **[FD]** What is Functional Dependency?
- ■ **[FD]** What is Partial Dependency?
- ■ **[FD]** What is Transitive Dependency?
- ■ **[ER Model]** Entities vs Attributes vs Relationships in ER
- ■ **[ER Model]** Weak Entity vs Strong Entity
- ■ **[ER Model]** Cardinality in ER (1:1, 1:N, M:N)
- ■ **[ER Model]** Convert ER Diagram to Relational Schema
- ■ **[Indexing]** What is B-Tree Index?
- ■ **[Indexing]** What is B+ Tree? Advantages over B-Tree?
- ■ **[Views]** What is a View? Advantages?
- ■ **[Views]** Difference between View and Table
- ■ **[Relational Algebra]** What is Relational Algebra? Operators.
- ■ **[Relational Algebra]** Selection vs Projection in Relational Algebra
- ■ **[Relational Algebra]** What is a Join in Relational Algebra?

■ **[Advanced]** OLTP vs OLAP	■ **[Recovery]** What is Recovery in DBMS? Techniques?
■ **[SQL-DBMS]** What is a Self-Join?	■ **[Keys]** What is a Surrogate Key?
■ **[Transactions]** States of a Transaction (Active, Partially Committed, etc.)	■ **[Concurrency]** What is Two-Phase Locking (2PL)?
■ **[Normalization]** Explain 4NF	■ **[FD]** Trivial vs Non-trivial Functional Dependency
■ **[FD]** What is Full Functional Dependency?	■ **[FD]** What is Closure of Attribute Set?
■ **[ER Model]** Participation constraints (total vs partial)	■ **[Indexing]** Dense Index vs Sparse Index
■ **[Indexing]** What is a Hash Index?	■ **[Procedures]** What is a Stored Procedure?
■ **[Procedures]** Stored Procedure vs Function	■ **[Triggers]** What is a Trigger?
■ **[Cursors]** What is a Cursor?	■ **[Query Processing]** What is Query Optimization?
■ **[Query Processing]** What is a Query Execution Plan?	■ **[Advanced]** What is a Data Warehouse?
■ **[Distributed DB]** What is Sharding?	■ **[Distributed DB]** What is Replication in databases?
■ **[Distributed DB]** CAP Theorem	■ **[NoSQL]** What is NoSQL? Types of NoSQL databases?
■ **[NoSQL]** SQL vs NoSQL – when to use which?	■ **[Recovery]** Write-Ahead Logging (WAL)
■ **[Recovery]** What is a Checkpoint in DBMS?	■ **[Concurrency]** What is Optimistic Concurrency Control?
■ **[Concurrency]** What is Timestamp Ordering protocol?	■ **[Normalization]** Explain 5NF
■ **[FD]** What is a Multi-valued Dependency?	

SECTION 4 OS — Operating Systems

Basics (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is an Operating System? Functions?	Basics	Very High	NQT/Digital	Interview
2	Types of Operating Systems	Basics	High	NQT	Interview
3	What is a Kernel? Types of kernel?	Basics	High	NQT/Digital	Interview
4	Monolithic vs Microkernel	Basics	Medium	Digital	Interview
5	Boot Process of OS	Basics	Medium	Digital	Interview

Processes (4 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Process? Process vs Program?	Processes	Very High	NQT/Digital	Interview
2	Process States and State Diagram	Processes	Very High	NQT/Digital	Interview
3	What is a PCB (Process Control Block)?	Processes	High	Digital	Interview
4	What is Context Switching?	Processes	Very High	Digital/Prime	Interview

Threads (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Thread?	Threads	Very High	NQT/Digital	Interview
2	Difference between Process and Thread	Threads	Very High	NQT/Digital/Prime	Interview
3	User-level Threads vs Kernel-level Threads	Threads	High	Digital	Interview
4	Benefits of multithreading	Threads	High	NQT/Digital	Interview
5	What is Concurrency vs Parallelism?	Threads	High	Digital/Prime	Interview

Scheduling (10 questions)

#	Question	Topic	Freq	Round	Interview
1	CPU Scheduling – what is it?	Scheduling	Very High	NQT/Digital	Interview
2	FCFS Scheduling – explain with example	Scheduling	Very High	NQT/Digital	Interview
3	SJF Scheduling – preemptive and non-preemptive	Scheduling	Very High	NQT/Digital	Interview
4	Round Robin Scheduling	Scheduling	Very High	NQT/Digital	Interview
5	Priority Scheduling	Scheduling	High	NQT/Digital	Interview
6	Multilevel Queue Scheduling	Scheduling	Medium	Digital	Interview

#	Question	Topic	Freq	Round	Interview
7	Multilevel Feedback Queue	Scheduling	Medium	Digital/Prime	Interview
8	Calculate average waiting time and turnaround time	Scheduling	Very High	NQT/Digital	Interview
9	What is CPU Burst and I/O Burst?	Scheduling	High	NQT/Digital	Interview
10	What is Preemptive vs Non-Preemptive scheduling?	Scheduling	High	NQT/Digital	Interview

Deadlocks (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Deadlock?	Deadlocks	Very High	Digital/Prime	Interview
2	Necessary conditions for Deadlock (Coffman)	Deadlocks	Very High	Digital/Prime	Interview
3	Deadlock Prevention	Deadlocks	High	Digital/Prime	Interview
4	Deadlock Avoidance – Banker's Algorithm	Deadlocks	High	Digital/Prime	Interview
5	Deadlock Detection and Recovery	Deadlocks	High	Digital/Prime	Interview
6	What is Starvation? Vs Deadlock?	Deadlocks	High	Digital	Interview
7	What is Livelock?	Deadlocks	Medium	Prime	Interview

Synchronization (11 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Semaphore?	Synchronization	Very High	Digital/Prime	Interview
2	Binary Semaphore vs Counting Semaphore	Synchronization	High	Digital/Prime	Interview
3	What is a Mutex?	Synchronization	Very High	Digital/Prime	Interview
4	Mutex vs Semaphore	Synchronization	Very High	Digital/Prime	Interview
5	What is a Monitor?	Synchronization	Medium	Digital	Interview
6	Producer-Consumer Problem	Synchronization	Very High	Digital/Prime	Interview
7	Readers-Writers Problem	Synchronization	High	Digital/Prime	Interview
8	Dining Philosophers Problem	Synchronization	High	Digital/Prime	Interview
9	What is Critical Section? Requirements?	Synchronization	Very High	Digital/Prime	Interview
10	Peterson's Solution	Synchronization	Medium	Digital	Interview
11	Test-and-Set, Compare-and-Swap	Synchronization	Medium	Digital	Interview

Memory (19 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Memory Management?	Memory	Very High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
2	Contiguous vs Non-Contiguous Memory Allocation	Memory	High	NQT/Digital	Interview
3	What is Fragmentation? Internal vs External?	Memory	Very High	NQT/Digital	Interview
4	What is Paging?	Memory	Very High	NQT/Digital	Interview
5	What is Segmentation?	Memory	High	NQT/Digital	Interview
6	Paging vs Segmentation	Memory	Very High	NQT/Digital	Interview
7	What is a Page Table?	Memory	High	NQT/Digital	Interview
8	Multilevel Page Table	Memory	Medium	Digital	Interview
9	What is TLB (Translation Lookaside Buffer)?	Memory	High	Digital/Prime	Interview
10	What is Virtual Memory?	Memory	Very High	Digital/Prime	Interview
11	What is Demand Paging?	Memory	High	Digital/Prime	Interview
12	What is a Page Fault?	Memory	Very High	Digital/Prime	Interview
13	Page Replacement Algorithms – FIFO	Memory	Very High	NQT/Digital	Interview
14	Page Replacement – Optimal (OPT)	Memory	High	Digital	Interview
15	Page Replacement – LRU	Memory	Very High	Digital/Prime	Interview
16	Page Replacement – LFU, MFU, Clock	Memory	Medium	Digital	Interview
17	Calculate number of page faults given reference string	Memory	Very High	NQT/Digital	Interview
18	What is Thrashing?	Memory	High	Digital/Prime	Interview
19	What is Swapping?	Memory	High	NQT/Digital	Interview

File System (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a File System?	File System	High	NQT/Digital	Interview
2	File Allocation Methods – Contiguous, Linked, Indexed	File System	High	Digital	Interview
3	FAT, Inode file system structure	File System	Medium	Digital	Interview
4	What is a Directory?	File System	Medium	NQT	Interview
5	Absolute vs Relative Path	File System	High	NQT	Interview

System Calls (4 questions)

#	Question	Topic	Freq	Round	Interview
1	What are System Calls?	System Calls	High	Digital/Prime	Interview
2	Types of System Calls	System Calls	High	Digital	Interview
3	fork(), exec(), wait() in Unix	System Calls	High	Digital/Prime	Interview
4	Zombie Process vs Orphan Process	System Calls	High	Digital/Prime	Interview

I/O (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Spooling?	I/O	Medium	NQT	Interview
2	What is Buffering?	I/O	Medium	NQT	Interview
3	What is Caching?	I/O	High	NQT/Digital	Interview
4	Cache Memory – L1, L2, L3	I/O	High	NQT/Digital	Interview
5	What is Interrupt? Types?	I/O	High	Digital	Interview
6	DMA (Direct Memory Access)	I/O	Medium	Digital	Interview
7	Disk Scheduling – FCFS, SSTF, SCAN, C-SCAN, LOOK	I/O	High	Digital/Prime	Interview

TOP 25 — OS — Operating Systems

#	Question	Topic	Freq	Round	Interview
1	What is an Operating System? Functions?	Basics	Very High	NQT/Digital	Interview
2	What is a Process? Process vs Program?	Processes	Very High	NQT/Digital	Interview
3	Process States and State Diagram	Processes	Very High	NQT/Digital	Interview
4	What is Context Switching?	Processes	Very High	Digital/Prime	Interview
5	What is a Thread?	Threads	Very High	NQT/Digital	Interview
6	Difference between Process and Thread	Threads	Very High	NQT/Digital/Prime	Interview
7	CPU Scheduling – what is it?	Scheduling	Very High	NQT/Digital	Interview
8	FCFS Scheduling – explain with example	Scheduling	Very High	NQT/Digital	Interview
9	SJF Scheduling – preemptive and non-preemptive	Scheduling	Very High	NQT/Digital	Interview
10	Round Robin Scheduling	Scheduling	Very High	NQT/Digital	Interview
11	Calculate average waiting time and turnaround time	Scheduling	Very High	NQT/Digital	Interview
12	What is a Deadlock?	Deadlocks	Very High	Digital/Prime	Interview
13	Necessary conditions for Deadlock (Coffman)	Deadlocks	Very High	Digital/Prime	Interview
14	What is a Semaphore?	Synchronization	Very High	Digital/Prime	Interview
15	What is a Mutex?	Synchronization	Very High	Digital/Prime	Interview
16	Mutex vs Semaphore	Synchronization	Very High	Digital/Prime	Interview
17	Producer-Consumer Problem	Synchronization	Very High	Digital/Prime	Interview
18	What is Critical Section? Requirements?	Synchronization	Very High	Digital/Prime	Interview
19	What is Memory Management?	Memory	Very High	NQT/Digital	Interview
20	What is Fragmentation? Internal vs External?	Memory	Very High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
21	What is Paging?	Memory	Very High	NQT/Digital	Interview
22	Paging vs Segmentation	Memory	Very High	NQT/Digital	Interview
23	What is Virtual Memory?	Memory	Very High	Digital/Prime	Interview
24	What is a Page Fault?	Memory	Very High	Digital/Prime	Interview
25	Page Replacement Algorithms – FIFO	Memory	Very High	NQT/Digital	Interview

TOP 50 — OS — Operating Systems

#	Question	Topic	Freq	Round	Interview
1	What is an Operating System? Functions?	Basics	Very High	NQT/Digital	Interview
2	What is a Process? Process vs Program?	Processes	Very High	NQT/Digital	Interview
3	Process States and State Diagram	Processes	Very High	NQT/Digital	Interview
4	What is Context Switching?	Processes	Very High	Digital/Prime	Interview
5	What is a Thread?	Threads	Very High	NQT/Digital	Interview
6	Difference between Process and Thread	Threads	Very High	NQT/Digital/Prime	Interview
7	CPU Scheduling – what is it?	Scheduling	Very High	NQT/Digital	Interview
8	FCFS Scheduling – explain with example	Scheduling	Very High	NQT/Digital	Interview
9	SJF Scheduling – preemptive and non-preemptive	Scheduling	Very High	NQT/Digital	Interview
10	Round Robin Scheduling	Scheduling	Very High	NQT/Digital	Interview
11	Calculate average waiting time and turnaround time	Scheduling	Very High	NQT/Digital	Interview
12	What is a Deadlock?	Deadlocks	Very High	Digital/Prime	Interview
13	Necessary conditions for Deadlock (Coffman)	Deadlocks	Very High	Digital/Prime	Interview
14	What is a Semaphore?	Synchronization	Very High	Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
15	What is a Mutex?	Synchronization	Very High	Digital/Prime	Interview
16	Mutex vs Semaphore	Synchronization	Very High	Digital/Prime	Interview
17	Producer-Consumer Problem	Synchronization	Very High	Digital/Prime	Interview
18	What is Critical Section? Requirements?	Synchronization	Very High	Digital/Prime	Interview
19	What is Memory Management?	Memory	Very High	NQT/Digital	Interview
20	What is Fragmentation? Internal vs External?	Memory	Very High	NQT/Digital	Interview
21	What is Paging?	Memory	Very High	NQT/Digital	Interview
22	Paging vs Segmentation	Memory	Very High	NQT/Digital	Interview
23	What is Virtual Memory?	Memory	Very High	Digital/Prime	Interview
24	What is a Page Fault?	Memory	Very High	Digital/Prime	Interview
25	Page Replacement Algorithms – FIFO	Memory	Very High	NQT/Digital	Interview
26	Page Replacement – LRU	Memory	Very High	Digital/Prime	Interview
27	Calculate number of page faults given reference string	Memory	Very High	NQT/Digital	Interview
28	Types of Operating Systems	Basics	High	NQT	Interview
29	What is a Kernel? Types of kernel?	Basics	High	NQT/Digital	Interview
30	What is a PCB (Process Control Block)?	Processes	High	Digital	Interview
31	User-level Threads vs Kernel-level Threads	Threads	High	Digital	Interview
32	Benefits of multithreading	Threads	High	NQT/Digital	Interview
33	What is Concurrency vs Parallelism?	Threads	High	Digital/Prime	Interview
34	Priority Scheduling	Scheduling	High	NQT/Digital	Interview
35	What is CPU Burst and I/O Burst?	Scheduling	High	NQT/Digital	Interview
36	What is Preemptive vs Non-Preemptive scheduling?	Scheduling	High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
37	Deadlock Prevention	Deadlocks	High	Digital/Prime	Interview
38	Deadlock Avoidance – Banker's Algorithm	Deadlocks	High	Digital/Prime	Interview
39	Deadlock Detection and Recovery	Deadlocks	High	Digital/Prime	Interview
40	What is Starvation? Vs Deadlock?	Deadlocks	High	Digital	Interview
41	Binary Semaphore vs Counting Semaphore	Synchronization	High	Digital/Prime	Interview
42	Readers-Writers Problem	Synchronization	High	Digital/Prime	Interview
43	Dining Philosophers Problem	Synchronization	High	Digital/Prime	Interview
44	Contiguous vs Non-Contiguous Memory Allocation	Memory	High	NQT/Digital	Interview
45	What is Segmentation?	Memory	High	NQT/Digital	Interview
46	What is a Page Table?	Memory	High	NQT/Digital	Interview
47	What is TLB (Translation Lookaside Buffer)?	Memory	High	Digital/Prime	Interview
48	What is Demand Paging?	Memory	High	Digital/Prime	Interview
49	Page Replacement – Optimal (OPT)	Memory	High	Digital	Interview
50	What is Thrashing?	Memory	High	Digital/Prime	Interview

MASTER OS — OPERATING SYSTEMS CHECKLIST

- ■ **[Basics]** What is an Operating System? Functions?
- ■ **[Processes]** What is a Process? Process vs Program?
- ■ **[Processes]** Process States and State Diagram
- ■ **[Processes]** What is Context Switching?
- ■ **[Threads]** What is a Thread?
- ■ **[Threads]** Difference between Process and Thread
- ■ **[Scheduling]** CPU Scheduling – what is it?
- ■ **[Scheduling]** FCFS Scheduling – explain with example
- ■ **[Scheduling]** SJF Scheduling – preemptive and non-preemptive
- ■ **[Scheduling]** Round Robin Scheduling
- ■ **[Scheduling]** Calculate average waiting time and turnaround time
- ■ **[Deadlocks]** What is a Deadlock?
- ■ **[Deadlocks]** Necessary conditions for Deadlock (Coffman)
- ■ **[Synchronization]** What is a Semaphore?
- ■ **[Synchronization]** What is a Mutex?
- ■ **[Synchronization]** Mutex vs Semaphore
- ■ **[Synchronization]** Producer-Consumer Problem
- ■ **[Synchronization]** What is Critical Section? Requirements?
- ■ **[Memory]** What is Memory Management?
- ■ **[Memory]** What is Fragmentation? Internal vs External?
- ■ **[Memory]** What is Paging?
- ■ **[Memory]** Paging vs Segmentation
- ■ **[Memory]** What is Virtual Memory?
- ■ **[Memory]** What is a Page Fault?
- ■ **[Memory]** Page Replacement Algorithms – FIFO
- ■ **[Memory]** Page Replacement – LRU
- ■ **[Memory]** Calculate number of page faults given reference string
- ■ **[Basics]** Types of Operating Systems
- ■ **[Basics]** What is a Kernel? Types of kernel?
- ■ **[Processes]** What is a PCB (Process Control Block)?
- ■ **[Threads]** User-level Threads vs Kernel-level Threads
- ■ **[Threads]** Benefits of multithreading
- ■ **[Threads]** What is Concurrency vs Parallelism?
- ■ **[Scheduling]** Priority Scheduling
- ■ **[Scheduling]** What is CPU Burst and I/O Burst?
- ■ **[Scheduling]** What is Preemptive vs Non-Preemptive scheduling?
- ■ **[Deadlocks]** Deadlock Prevention
- ■ **[Deadlocks]** Deadlock Avoidance – Banker's Algorithm
- ■ **[Deadlocks]** Deadlock Detection and Recovery
- ■ **[Deadlocks]** What is Starvation? Vs Deadlock?
- ■ **[Synchronization]** Binary Semaphore vs Counting Semaphore
- ■ **[Synchronization]** Readers-Writers Problem
- ■ **[Synchronization]** Dining Philosophers Problem
- ■ **[Memory]** Contiguous vs Non-Contiguous Memory Allocation
- ■ **[Memory]** What is Segmentation?
- ■ **[Memory]** What is a Page Table?
- ■ **[Memory]** What is TLB (Translation Lookaside Buffer)?
- ■ **[Memory]** What is Demand Paging?
- ■ **[Memory]** Page Replacement – Optimal (OPT)
- ■ **[Memory]** What is Thrashing?
- ■ **[File System]** What is a File System?
- ■ **[File System]** File Allocation Methods – Contiguous, Linked, Indexed
- ■ **[File System]** Absolute vs Relative Path
- ■ **[System Calls]** What are System Calls?
- ■ **[System Calls]** Types of System Calls
- ■ **[System Calls]** fork(), exec(), wait() in Unix

■ **[System Calls]** Zombie Process vs Orphan Process	■ **[I/O]** What is Caching?
■ **[I/O]** Cache Memory – L1, L2, L3	■ **[I/O]** What is Interrupt? Types?
■ **[I/O]** Disk Scheduling – FCFS, SSTF, SCAN, C-SCAN, LOOK	■ **[Memory]** What is Swapping?
■ **[Basics]** Monolithic vs Microkernel	■ **[Scheduling]** Multilevel Queue Scheduling
■ **[Scheduling]** Multilevel Feedback Queue	■ **[Deadlocks]** What is Livelock?
■ **[Synchronization]** What is a Monitor?	■ **[Synchronization]** Peterson's Solution
■ **[Synchronization]** Test-and-Set, Compare-and-Swap	■ **[Memory]** Multilevel Page Table
■ **[Memory]** Page Replacement – LFU, MFU, Clock	■ **[File System]** FAT, Inode file system structure
■ **[File System]** What is a Directory?	■ **[I/O]** What is Spooling?
■ **[I/O]** What is Buffering?	■ **[I/O]** DMA (Direct Memory Access)
■ **[Basics]** Boot Process of OS	

SECTION 5 CN — Computer Networks

Basics (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Computer Network? Types?	Basics	Very High	NQT	Interview
2	What is a Protocol?	Basics	High	NQT	Interview
3	LAN vs WAN vs MAN	Basics	High	NQT	Interview
4	What is Bandwidth? Throughput? Latency?	Basics	High	NQT/Digital	Interview
5	Network Topology types – Star, Bus, Ring, Mesh	Basics	High	NQT	Interview

OSI (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is the OSI Model? Explain 7 layers.	OSI	Very High	NQT/Digital/Prime	Interview
2	Function of each OSI layer	OSI	Very High	NQT/Digital	Interview
3	OSI vs TCP/IP model	OSI	Very High	NQT/Digital	Interview
4	What is the Physical Layer? Devices?	OSI	High	NQT	Interview
5	What is the Data Link Layer?	OSI	High	NQT/Digital	Interview
6	What is the Network Layer?	OSI	High	NQT/Digital	Interview
7	What is the Transport Layer?	OSI	High	NQT/Digital	Interview

TCP/IP (1 questions)

#	Question	Topic	Freq	Round	Interview
1	TCP/IP Model – 4 layers explained	TCP/IP	Very High	NQT/Digital	Interview

Data Link (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is MAC Address?	Data Link	High	NQT/Digital	Interview
2	Ethernet – CSMA/CD	Data Link	High	Digital	Interview
3	WiFi – CSMA/CA	Data Link	Medium	Digital	Interview
4	Stop-and-Wait vs Go-Back-N vs Selective Repeat	Data Link	High	Digital/Prime	Interview
5	What is Multiplexing? Types?	Data Link	Medium	Digital	Interview
6	What is a Broadcast Domain vs Collision Domain?	Data Link	Medium	Digital	Interview

IP (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is IP Address? IPv4 vs IPv6?	IP	Very High	NQT/Digital	Interview
2	What is Subnetting? CIDR notation?	IP	High	Digital/Prime	Interview
3	Public vs Private IP Address	IP	High	NQT/Digital	Interview
4	What is NAT (Network Address Translation)?	IP	High	Digital	Interview
5	What is IPv6? Advantages over IPv4?	IP	Medium	Digital	Interview
6	IPv4 Packet Header fields	IP	Medium	Digital/Prime	Interview

TCP/UDP (2 questions)

#	Question	Topic	Freq	Round	Interview
1	Difference between TCP and UDP	TCP/UDP	Very High	NQT/Digital/Prime	Interview
2	TCP vs UDP – when to use which?	TCP/UDP	High	Digital/Prime	Interview

TCP (6 questions)

#	Question	Topic	Freq	Round	Interview
1	TCP – Three-Way Handshake	TCP	Very High	NQT/Digital/Prime	Interview
2	TCP – Four-Way Termination	TCP	High	Digital/Prime	Interview
3	TCP Features – reliability, flow control, congestion	TCP	High	Digital/Prime	Interview
4	What is Congestion Control?	TCP	High	Digital/Prime	Interview
5	TCP Slow Start, Congestion Avoidance	TCP	Medium	Digital/Prime	Interview
6	What is Flow Control? Sliding Window?	TCP	High	Digital/Prime	Interview

Ports (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Port Number? Well-known ports?	Ports	High	NQT/Digital	Interview

DNS (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is DNS? How does it work?	DNS	Very High	NQT/Digital/Prime	Interview
2	DNS Resolution Process (step by step)	DNS	High	Digital/Prime	Interview
3	Types of DNS Records (A, AAAA, CNAME, MX, NS)	DNS	Medium	Digital	Interview

DHCP (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is DHCP? How does it work?	DHCP	High	NQT/Digital	Interview
2	DHCP DORA Process	DHCP	Medium	Digital	Interview

HTTP (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is HTTP? How does it work?	HTTP	Very High	NQT/Digital	Interview
2	HTTP vs HTTPS	HTTP	Very High	NQT/Digital/Prime	Interview
3	HTTP Methods – GET, POST, PUT, DELETE, PATCH	HTTP	High	Digital/Prime	Interview
4	HTTP Status Codes (200, 301, 404, 500 etc.)	HTTP	High	Digital	Interview
5	What is a Cookie? Session? Token?	HTTP	High	Digital/Prime	Interview

Security (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is SSL/TLS?	Security	High	Digital/Prime	Interview
2	What is a Firewall? Types?	Security	High	Digital/Prime	Interview
3	What is a Proxy Server?	Security	High	Digital	Interview
4	What is VPN?	Security	High	Digital/Prime	Interview
5	What is 3-Way Handshake for HTTPS?	Security	High	Digital/Prime	Interview

FTP (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is FTP? Active vs Passive mode?	FTP	Medium	Digital	Interview

Email (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is SMTP, POP3, IMAP?	Email	High	NQT/Digital	Interview
2	Difference between POP3 and IMAP	Email	High	NQT/Digital	Interview

Routing (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Router? How does routing work?	Routing	Very High	NQT/Digital	Interview
2	Static Routing vs Dynamic Routing	Routing	High	Digital	Interview
3	Routing Protocols – RIP, OSPF, BGP	Routing	High	Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
4	Distance Vector vs Link State routing	Routing	High	Digital/Prime	Interview
5	Dijkstra for Routing (OSPF)	Routing	Medium	Digital/Prime	Interview
6	What is Flooding in routing?	Routing	Medium	Digital	Interview
7	What is BGP (Border Gateway Protocol)?	Routing	Low	Prime	Interview

Switching (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Switch?	Switching	Very High	NQT/Digital	Interview

Network Devices (3 questions)

#	Question	Topic	Freq	Round	Interview
1	Difference between Hub, Switch, and Router	Network Devices	Very High	NQT/Digital/Prime	Interview
2	What is a Gateway?	Network Devices	High	NQT	Interview
3	What is a Bridge?	Network Devices	Medium	NQT	Interview

Protocols (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is ARP? How does it work?	Protocols	High	Digital	Interview
2	What is RARP?	Protocols	Medium	Digital	Interview
3	What is a Ping? How does it work (ICMP)?	Protocols	High	NQT/Digital	Interview
4	Traceroute – how does it work?	Protocols	Medium	Digital	Interview
5	What is ICMP?	Protocols	High	Digital	Interview
6	What is Tunneling?	Protocols	Low	Prime	Interview

LAN (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is VLAN?	LAN	Medium	Digital	Interview

Error Handling (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Error Detection? CRC, Checksum, Parity?	Error Handling	High	Digital	Interview
2	What is Hamming Code?	Error Handling	Medium	Digital	Interview

Sockets (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Socket? Socket programming basics?	Sockets	High	Digital/Prime	Interview

Web (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a CDN (Content Delivery Network)?	Web	Medium	Digital	Interview
2	What is Load Balancing?	Web	Medium	Digital/Prime	Interview

TOP 25 — CN — Computer Networks

#	Question	Topic	Freq	Round	Interview
1	What is a Computer Network? Types?	Basics	Very High	NQT	Interview
2	What is the OSI Model? Explain 7 layers.	OSI	Very High	NQT/Digital/Prime	Interview
3	Function of each OSI layer	OSI	Very High	NQT/Digital	Interview
4	OSI vs TCP/IP model	OSI	Very High	NQT/Digital	Interview
5	TCP/IP Model – 4 layers explained	TCP/IP	Very High	NQT/Digital	Interview
6	What is IP Address? IPv4 vs IPv6?	IP	Very High	NQT/Digital	Interview
7	Difference between TCP and UDP	TCP/UDP	Very High	NQT/Digital/Prime	Interview
8	TCP – Three-Way Handshake	TCP	Very High	NQT/Digital/Prime	Interview
9	What is DNS? How does it work?	DNS	Very High	NQT/Digital/Prime	Interview
10	What is HTTP? How does it work?	HTTP	Very High	NQT/Digital	Interview
11	HTTP vs HTTPS	HTTP	Very High	NQT/Digital/Prime	Interview
12	What is a Router? How does routing work?	Routing	Very High	NQT/Digital	Interview
13	What is a Switch?	Switching	Very High	NQT/Digital	Interview
14	Difference between Hub, Switch, and Router	Network Devices	Very High	NQT/Digital/Prime	Interview
15	What is a Protocol?	Basics	High	NQT	Interview
16	LAN vs WAN vs MAN	Basics	High	NQT	Interview
17	What is Bandwidth? Throughput? Latency?	Basics	High	NQT/Digital	Interview
18	What is the Physical Layer? Devices?	OSI	High	NQT	Interview
19	What is the Data Link Layer?	OSI	High	NQT/Digital	Interview
20	What is MAC Address?	Data Link	High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
21	What is the Network Layer?	OSI	High	NQT/Digital	Interview
22	What is Subnetting? CIDR notation?	IP	High	Digital/Prime	Interview
23	Public vs Private IP Address	IP	High	NQT/Digital	Interview
24	What is NAT (Network Address Translation)?	IP	High	Digital	Interview
25	What is the Transport Layer?	OSI	High	NQT/Digital	Interview

TOP 50 — CN — Computer Networks

#	Question	Topic	Freq	Round	Interview
1	What is a Computer Network? Types?	Basics	Very High	NQT	Interview
2	What is the OSI Model? Explain 7 layers.	OSI	Very High	NQT/Digital/Prime	Interview
3	Function of each OSI layer	OSI	Very High	NQT/Digital	Interview
4	OSI vs TCP/IP model	OSI	Very High	NQT/Digital	Interview
5	TCP/IP Model – 4 layers explained	TCP/IP	Very High	NQT/Digital	Interview
6	What is IP Address? IPv4 vs IPv6?	IP	Very High	NQT/Digital	Interview
7	Difference between TCP and UDP	TCP/UDP	Very High	NQT/Digital/Prime	Interview
8	TCP – Three-Way Handshake	TCP	Very High	NQT/Digital/Prime	Interview
9	What is DNS? How does it work?	DNS	Very High	NQT/Digital/Prime	Interview
10	What is HTTP? How does it work?	HTTP	Very High	NQT/Digital	Interview
11	HTTP vs HTTPS	HTTP	Very High	NQT/Digital/Prime	Interview
12	What is a Router? How does routing work?	Routing	Very High	NQT/Digital	Interview
13	What is a Switch?	Switching	Very High	NQT/Digital	Interview
14	Difference between Hub, Switch, and Router	Network Devices	Very High	NQT/Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
15	What is a Protocol?	Basics	High	NQT	Interview
16	LAN vs WAN vs MAN	Basics	High	NQT	Interview
17	What is Bandwidth? Throughput? Latency?	Basics	High	NQT/Digital	Interview
18	What is the Physical Layer? Devices?	OSI	High	NQT	Interview
19	What is the Data Link Layer?	OSI	High	NQT/Digital	Interview
20	What is MAC Address?	Data Link	High	NQT/Digital	Interview
21	What is the Network Layer?	OSI	High	NQT/Digital	Interview
22	What is Subnetting? CIDR notation?	IP	High	Digital/Prime	Interview
23	Public vs Private IP Address	IP	High	NQT/Digital	Interview
24	What is NAT (Network Address Translation)?	IP	High	Digital	Interview
25	What is the Transport Layer?	OSI	High	NQT/Digital	Interview
26	TCP – Four-Way Termination	TCP	High	Digital/Prime	Interview
27	TCP Features – reliability, flow control, congestion	TCP	High	Digital/Prime	Interview
28	TCP vs UDP – when to use which?	TCP/UDP	High	Digital/Prime	Interview
29	What is a Port Number? Well-known ports?	Ports	High	NQT/Digital	Interview
30	DNS Resolution Process (step by step)	DNS	High	Digital/Prime	Interview
31	What is DHCP? How does it work?	DHCP	High	NQT/Digital	Interview
32	HTTP Methods – GET, POST, PUT, DELETE, PATCH	HTTP	High	Digital/Prime	Interview
33	HTTP Status Codes (200, 301, 404, 500 etc.)	HTTP	High	Digital	Interview
34	What is SSL/TLS?	Security	High	Digital/Prime	Interview
35	What is a Cookie? Session? Token?	HTTP	High	Digital/Prime	Interview
36	What is SMTP, POP3, IMAP?	Email	High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
37	Difference between POP3 and IMAP	Email	High	NQT/Digital	Interview
38	What is a Gateway?	Network Devices	High	NQT	Interview
39	What is ARP? How does it work?	Protocols	High	Digital	Interview
40	Static Routing vs Dynamic Routing	Routing	High	Digital	Interview
41	Routing Protocols – RIP, OSPF, BGP	Routing	High	Digital/Prime	Interview
42	Distance Vector vs Link State routing	Routing	High	Digital/Prime	Interview
43	Ethernet – CSMA/CD	Data Link	High	Digital	Interview
44	What is Congestion Control?	TCP	High	Digital/Prime	Interview
45	What is Flow Control? Sliding Window?	TCP	High	Digital/Prime	Interview
46	Stop-and-Wait vs Go-Back-N vs Selective Repeat	Data Link	High	Digital/Prime	Interview
47	What is Error Detection? CRC, Checksum, Parity?	Error Handling	High	Digital	Interview
48	What is a Firewall? Types?	Security	High	Digital/Prime	Interview
49	What is a Proxy Server?	Security	High	Digital	Interview
50	What is VPN?	Security	High	Digital/Prime	Interview

MASTER CN — COMPUTER NETWORKS CHECKLIST

■ **[Basics]** What is a Computer Network? Types?	■ **[OSI]** What is the OSI Model? Explain 7 layers.
■ **[OSI]** Function of each OSI layer	■ **[OSI]** OSI vs TCP/IP model
■ **[TCP/IP]** TCP/IP Model – 4 layers explained	■ **[IP]** What is IP Address? IPv4 vs IPv6?
■ **[TCP/UDP]** Difference between TCP and UDP	■ **[TCP]** TCP – Three-Way Handshake
■ **[DNS]** What is DNS? How does it work?	■ **[HTTP]** What is HTTP? How does it work?
■ **[HTTP]** HTTP vs HTTPS	■ **[Routing]** What is a Router? How does routing work?
■ **[Switching]** What is a Switch?	■ **[Network Devices]** Difference between Hub, Switch, and Router
■ **[Basics]** What is a Protocol?	■ **[Basics]** LAN vs WAN vs MAN
■ **[Basics]** What is Bandwidth? Throughput? Latency?	■ **[OSI]** What is the Physical Layer? Devices?
■ **[OSI]** What is the Data Link Layer?	■ **[Data Link]** What is MAC Address?
■ **[OSI]** What is the Network Layer?	■ **[IP]** What is Subnetting? CIDR notation?
■ **[IP]** Public vs Private IP Address	■ **[IP]** What is NAT (Network Address Translation)?
■ **[OSI]** What is the Transport Layer?	■ **[TCP]** TCP – Four-Way Termination
■ **[TCP]** TCP Features – reliability, flow control, congestion	■ **[TCP/UDP]** TCP vs UDP – when to use which?
■ **[Ports]** What is a Port Number? Well-known ports?	■ **[DNS]** DNS Resolution Process (step by step)
■ **[DHCP]** What is DHCP? How does it work?	■ **[HTTP]** HTTP Methods – GET, POST, PUT, DELETE, PATCH
■ **[HTTP]** HTTP Status Codes (200, 301, 404, 500 etc.)	■ **[Security]** What is SSL/TLS?
■ **[HTTP]** What is a Cookie? Session? Token?	■ **[Email]** What is SMTP, POP3, IMAP?
■ **[Email]** Difference between POP3 and IMAP	■ **[Network Devices]** What is a Gateway?
■ **[Protocols]** What is ARP? How does it work?	■ **[Routing]** Static Routing vs Dynamic Routing
■ **[Routing]** Routing Protocols – RIP, OSPF, BGP	■ **[Routing]** Distance Vector vs Link State routing
■ **[Data Link]** Ethernet – CSMA/CD	■ **[TCP]** What is Congestion Control?
■ **[TCP]** What is Flow Control? Sliding Window?	■ **[Data Link]** Stop-and-Wait vs Go-Back-N vs Selective Repeat
■ **[Error Handling]** What is Error Detection? CRC, Checksum, Parity?	■ **[Security]** What is a Firewall? Types?
■ **[Security]** What is a Proxy Server?	■ **[Security]** What is VPN?
■ **[Sockets]** What is a Socket? Socket programming basics?	■ **[Protocols]** What is a Ping? How does it work (ICMP)?
■ **[Protocols]** What is ICMP?	■ **[Basics]** Network Topology types – Star, Bus, Ring, Mesh
■ **[Security]** What is 3-Way Handshake for HTTPS?	■ **[DNS]** Types of DNS Records (A, AAAA, CNAME, MX, NS)
■ **[DHCP]** DHCP DORA Process	■ **[FTP]** What is FTP? Active vs Passive mode?

■ **[Network Devices]** What is a Bridge?	■ **[Protocols]** What is RARP?
■ **[Routing]** Dijkstra for Routing (OSPF)	■ **[Routing]** What is Flooding in routing?
■ **[LAN]** What is VLAN?	■ **[Data Link]** WiFi – CSMA/CA
■ **[TCP]** TCP Slow Start, Congestion Avoidance	■ **[Error Handling]** What is Hamming Code?
■ **[Web]** What is a CDN (Content Delivery Network)?	■ **[Web]** What is Load Balancing?
■ **[Data Link]** What is Multiplexing? Types?	■ **[Protocols]** Traceroute – how does it work?
■ **[IP]** What is IPv6? Advantages over IPv4?	■ **[IP]** IPv4 Packet Header fields
■ **[Data Link]** What is a Broadcast Domain vs Collision Domain?	■ **[Protocols]** What is Tunneling?
■ **[Routing]** What is BGP (Border Gateway Protocol)?	

SECTION 6 OOPs — Object-Oriented Programming

Basics (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is OOP? Advantages over procedural?	Basics	Very High	NQT/Digital	Interview
2	What is a Class? What is an Object?	Basics	Very High	NQT/Digital	Interview
3	Difference between Class and Object	Basics	Very High	NQT/Digital	Interview

Encapsulation (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Encapsulation?	Encapsulation	Very High	NQT/Digital/Prime	Interview

Abstraction (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Abstraction?	Abstraction	Very High	NQT/Digital/Prime	Interview

Core Concepts (1 questions)

#	Question	Topic	Freq	Round	Interview
1	Difference between Encapsulation and Abstraction	Core Concepts	Very High	NQT/Digital	Interview

Inheritance (4 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Inheritance?	Inheritance	Very High	NQT/Digital/Prime	Interview
2	Types of Inheritance (single, multiple, multilevel, hybrid)	Inheritance	Very High	NQT/Digital	Interview
3	What is Multiple Inheritance? Problems?	Inheritance	High	Digital/Prime	Interview
4	Diamond Problem in Multiple Inheritance	Inheritance	High	Digital/Prime	Interview

Polymorphism (9 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Polymorphism?	Polymorphism	Very High	NQT/Digital/Prime	Interview
2	Compile-time vs Runtime Polymorphism	Polymorphism	Very High	NQT/Digital/Prime	Interview
3	What is Method Overloading?	Polymorphism	Very High	NQT/Digital	Interview
4	What is Method Overriding?	Polymorphism	Very High	NQT/Digital	Interview
5	Difference between Overloading and Overriding	Polymorphism	Very High	NQT/Digital/Prime	Interview
6	What is operator overloading?	Polymorphism	High	Digital	Interview
7	Can we override static methods?	Polymorphism	High	Digital	Interview
8	Can we override private methods?	Polymorphism	High	Digital	Interview
9	Difference between method hiding and method overriding	Polymorphism	High	Digital	Interview

Constructors (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Constructor? Types?	Constructors	Very High	NQT/Digital	Interview
2	Default Constructor vs Parameterized Constructor	Constructors	Very High	NQT/Digital	Interview
3	What is a Copy Constructor?	Constructors	High	Digital	Interview
4	What is a Destructor?	Constructors	High	NQT/Digital	Interview
5	What is Constructor Overloading?	Constructors	High	NQT/Digital	Interview
6	Can we call a virtual function from constructor?	Constructors	Medium	Digital/Prime	Interview

Virtual (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Virtual Function?	Virtual	Very High	Digital/Prime	Interview
2	What is a Pure Virtual Function?	Virtual	High	Digital/Prime	Interview
3	What is function overriding with virtual table (vtable)?	Virtual	High	Digital/Prime	Interview
4	What is a vtable / vptr?	Virtual	High	Digital/Prime	Interview
5	What is dynamic dispatch?	Virtual	High	Digital/Prime	Interview
6	What is early binding vs late binding?	Virtual	High	Digital/Prime	Interview

Abstract (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is an Abstract Class?	Abstract	Very High	Digital/Prime	Interview
2	Difference between Abstract Class and Interface	Abstract	Very High	Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
3	Can we instantiate an Abstract Class?	Abstract	High	Digital	Interview

Interface (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is an Interface?	Interface	Very High	Digital/Prime	Interview
2	Can Interface have a constructor?	Interface	Medium	Digital	Interview
3	Difference between Interface and Concrete Class	Interface	High	Digital	Interview

Keywords (5 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Final/Sealed class?	Keywords	High	Digital	Interview
2	What does the 'static' keyword mean in OOP?	Keywords	Very High	NQT/Digital	Interview
3	What is a static method vs instance method?	Keywords	Very High	NQT/Digital	Interview
4	What is the 'this' keyword?	Keywords	Very High	NQT/Digital	Interview
5	What is the 'super' keyword?	Keywords	High	NQT/Digital	Interview

Access Modifiers (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What are Access Modifiers? (public, private, protected, default)	Access Modifiers	Very High	NQT/Digital	Interview
2	What is friend function in C++?	Access Modifiers	Medium	Digital	Interview

Generics (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a template/generic in OOP?	Generics	High	Digital/Prime	Interview

Exception (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Exception Handling? try-catch-finally?	Exception	Very High	NQT/Digital	Interview
2	Checked vs Unchecked Exceptions	Exception	High	Digital	Interview
3	What is a Custom Exception?	Exception	Medium	Digital	Interview

Memory (3 questions)

#	Question	Topic	Freq	Round	Interview
1	What is garbage collection?	Memory	High	Digital/Prime	Interview
2	What is a memory leak?	Memory	High	Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
3	What is shallow copy vs deep copy?	Memory	High	Digital/Prime	Interview

Relationships (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Association, Aggregation, Composition?	Relationships	High	Digital/Prime	Interview
2	Difference between Composition and Aggregation	Relationships	High	Digital/Prime	Interview

Design (7 questions)

#	Question	Topic	Freq	Round	Interview
1	What is Cohesion and Coupling?	Design	High	Digital/Prime	Interview
2	SOLID Principles – explain each	Design	High	Digital/Prime	Interview
3	Single Responsibility Principle	Design	High	Digital/Prime	Interview
4	Open/Closed Principle	Design	High	Digital/Prime	Interview
5	Liskov Substitution Principle	Design	High	Digital/Prime	Interview
6	Interface Segregation Principle	Design	Medium	Prime	Interview
7	Dependency Inversion Principle	Design	Medium	Prime	Interview

Design Patterns (6 questions)

#	Question	Topic	Freq	Round	Interview
1	What is a Design Pattern? Name common ones.	Design Patterns	High	Digital/Prime	Interview
2	Singleton Design Pattern	Design Patterns	High	Digital/Prime	Interview
3	Factory Design Pattern	Design Patterns	High	Digital/Prime	Interview
4	Observer Design Pattern	Design Patterns	Medium	Prime	Interview
5	Decorator Design Pattern	Design Patterns	Low	Prime	Interview
6	Strategy Design Pattern	Design Patterns	Low	Prime	Interview

Architecture (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is MVC pattern?	Architecture	High	Digital/Prime	Interview

OOP Concepts (1 questions)

#	Question	Topic	Freq	Round	Interview
1	What is the difference between IS-A and HAS-A relationship?	OOP Concepts	High	Digital	Interview

Advanced OOP (2 questions)

#	Question	Topic	Freq	Round	Interview
1	What is an inner class / nested class?	Advanced OOP	Medium	Digital	Interview
2	What is an anonymous class?	Advanced OOP	Medium	Digital	Interview

TOP 25 — OOPs — Object-Oriented Programming

#	Question	Topic	Freq	Round	Interview
1	What is OOP? Advantages over procedural?	Basics	Very High	NQT/Digital	Interview
2	What is a Class? What is an Object?	Basics	Very High	NQT/Digital	Interview
3	Difference between Class and Object	Basics	Very High	NQT/Digital	Interview
4	What is Encapsulation?	Encapsulation	Very High	NQT/Digital/Prime	Interview
5	What is Abstraction?	Abstraction	Very High	NQT/Digital/Prime	Interview
6	Difference between Encapsulation and Abstraction	Core Concepts	Very High	NQT/Digital	Interview
7	What is Inheritance?	Inheritance	Very High	NQT/Digital/Prime	Interview
8	Types of Inheritance (single, multiple, multilevel, hybrid)	Inheritance	Very High	NQT/Digital	Interview
9	What is Polymorphism?	Polymorphism	Very High	NQT/Digital/Prime	Interview
10	Compile-time vs Runtime Polymorphism	Polymorphism	Very High	NQT/Digital/Prime	Interview
11	What is Method Overloading?	Polymorphism	Very High	NQT/Digital	Interview
12	What is Method Overriding?	Polymorphism	Very High	NQT/Digital	Interview
13	Difference between Overloading and Overriding	Polymorphism	Very High	NQT/Digital/Prime	Interview
14	What is a Constructor? Types?	Constructors	Very High	NQT/Digital	Interview
15	Default Constructor vs Parameterized Constructor	Constructors	Very High	NQT/Digital	Interview
16	What is a Virtual Function?	Virtual	Very High	Digital/Prime	Interview
17	What is an Abstract Class?	Abstract	Very High	Digital/Prime	Interview
18	Difference between Abstract Class and Interface	Abstract	Very High	Digital/Prime	Interview
19	What is an Interface?	Interface	Very High	Digital/Prime	Interview
20	What does the 'static' keyword mean in OOP?	Keywords	Very High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
21	What is a static method vs instance method?	Keywords	Very High	NQT/Digital	Interview
22	What is the 'this' keyword?	Keywords	Very High	NQT/Digital	Interview
23	What are Access Modifiers? (public, private, protected, default)	Access Modifiers	Very High	NQT/Digital	Interview
24	What is Exception Handling? try-catch-finally?	Exception	Very High	NQT/Digital	Interview
25	What is Multiple Inheritance? Problems?	Inheritance	High	Digital/Prime	Interview

TOP 50 — OOPs — Object-Oriented Programming

#	Question	Topic	Freq	Round	Interview
1	What is OOP? Advantages over procedural?	Basics	Very High	NQT/Digital	Interview
2	What is a Class? What is an Object?	Basics	Very High	NQT/Digital	Interview
3	Difference between Class and Object	Basics	Very High	NQT/Digital	Interview
4	What is Encapsulation?	Encapsulation	Very High	NQT/Digital/Prime	Interview
5	What is Abstraction?	Abstraction	Very High	NQT/Digital/Prime	Interview
6	Difference between Encapsulation and Abstraction	Core Concepts	Very High	NQT/Digital	Interview
7	What is Inheritance?	Inheritance	Very High	NQT/Digital/Prime	Interview
8	Types of Inheritance (single, multiple, multilevel, hybrid)	Inheritance	Very High	NQT/Digital	Interview
9	What is Polymorphism?	Polymorphism	Very High	NQT/Digital/Prime	Interview
10	Compile-time vs Runtime Polymorphism	Polymorphism	Very High	NQT/Digital/Prime	Interview
11	What is Method Overloading?	Polymorphism	Very High	NQT/Digital	Interview
12	What is Method Overriding?	Polymorphism	Very High	NQT/Digital	Interview
13	Difference between Overloading and Overriding	Polymorphism	Very High	NQT/Digital/Prime	Interview
14	What is a Constructor? Types?	Constructors	Very High	NQT/Digital	Interview

#	Question	Topic	Freq	Round	Interview
15	Default Constructor vs Parameterized Constructor	Constructors	Very High	NQT/Digital	Interview
16	What is a Virtual Function?	Virtual	Very High	Digital/Prime	Interview
17	What is an Abstract Class?	Abstract	Very High	Digital/Prime	Interview
18	Difference between Abstract Class and Interface	Abstract	Very High	Digital/Prime	Interview
19	What is an Interface?	Interface	Very High	Digital/Prime	Interview
20	What does the 'static' keyword mean in OOP?	Keywords	Very High	NQT/Digital	Interview
21	What is a static method vs instance method?	Keywords	Very High	NQT/Digital	Interview
22	What is the 'this' keyword?	Keywords	Very High	NQT/Digital	Interview
23	What are Access Modifiers? (public, private, protected, default)	Access Modifiers	Very High	NQT/Digital	Interview
24	What is Exception Handling? try-catch-finally?	Exception	Very High	NQT/Digital	Interview
25	What is Multiple Inheritance? Problems?	Inheritance	High	Digital/Prime	Interview
26	Diamond Problem in Multiple Inheritance	Inheritance	High	Digital/Prime	Interview
27	What is a Copy Constructor?	Constructors	High	Digital	Interview
28	What is a Destructor?	Constructors	High	NQT/Digital	Interview
29	What is Constructor Overloading?	Constructors	High	NQT/Digital	Interview
30	What is a Pure Virtual Function?	Virtual	High	Digital/Prime	Interview
31	Can we instantiate an Abstract Class?	Abstract	High	Digital	Interview
32	Difference between Interface and Concrete Class	Interface	High	Digital	Interview
33	What is a Final/Sealed class?	Keywords	High	Digital	Interview
34	What is the 'super' keyword?	Keywords	High	NQT/Digital	Interview
35	What is operator overloading?	Polymorphism	High	Digital	Interview
36	What is function overriding with virtual table (vtable)?	Virtual	High	Digital/Prime	Interview

#	Question	Topic	Freq	Round	Interview
37	What is a vtable / vptr?	Virtual	High	Digital/Prime	Interview
38	What is dynamic dispatch?	Virtual	High	Digital/Prime	Interview
39	What is early binding vs late binding?	Virtual	High	Digital/Prime	Interview
40	What is a template/generic in OOP?	Generics	High	Digital/Prime	Interview
41	Checked vs Unchecked Exceptions	Exception	High	Digital	Interview
42	What is garbage collection?	Memory	High	Digital/Prime	Interview
43	What is a memory leak?	Memory	High	Digital/Prime	Interview
44	What is shallow copy vs deep copy?	Memory	High	Digital/Prime	Interview
45	What is Association, Aggregation, Composition?	Relationships	High	Digital/Prime	Interview
46	Difference between Composition and Aggregation	Relationships	High	Digital/Prime	Interview
47	What is Cohesion and Coupling?	Design	High	Digital/Prime	Interview
48	SOLID Principles – explain each	Design	High	Digital/Prime	Interview
49	Single Responsibility Principle	Design	High	Digital/Prime	Interview
50	Open/Closed Principle	Design	High	Digital/Prime	Interview

MASTER OOPS — OBJECT-ORIENTED PROGRAMMING CHECKLIST

■ **[Basics]** What is OOP? Advantages over procedural?	■ **[Basics]** What is a Class? What is an Object?
■ **[Basics]** Difference between Class and Object	■ **[Encapsulation]** What is Encapsulation?
■ **[Abstraction]** What is Abstraction?	■ **[Core Concepts]** Difference between Encapsulation and Abstraction
■ **[Inheritance]** What is Inheritance?	■ **[Inheritance]** Types of Inheritance (single, multiple, multilevel, hybrid)
■ **[Polymorphism]** What is Polymorphism?	■ **[Polymorphism]** Compile-time vs Runtime Polymorphism
■ **[Polymorphism]** What is Method Overloading?	■ **[Polymorphism]** What is Method Overriding?
■ **[Polymorphism]** Difference between Overloading and Overriding	■ **[Constructors]** What is a Constructor? Types?
■ **[Constructors]** Default Constructor vs Parameterized Constructor	■ **[Virtual]** What is a Virtual Function?
■ **[Abstract]** What is an Abstract Class?	■ **[Abstract]** Difference between Abstract Class and Interface
■ **[Interface]** What is an Interface?	■ **[Keywords]** What does the 'static' keyword mean in OOP?
■ **[Keywords]** What is a static method vs instance method?	■ **[Keywords]** What is the 'this' keyword?
■ **[Access Modifiers]** What are Access Modifiers? (public, private, protected, default)	■ **[Exception]** What is Exception Handling? try-catch-finally?
■ **[Inheritance]** What is Multiple Inheritance? Problems?	■ **[Inheritance]** Diamond Problem in Multiple Inheritance
■ **[Constructors]** What is a Copy Constructor?	■ **[Constructors]** What is a Destructor?
■ **[Constructors]** What is Constructor Overloading?	■ **[Virtual]** What is a Pure Virtual Function?
■ **[Abstract]** Can we instantiate an Abstract Class?	■ **[Interface]** Difference between Interface and Concrete Class
■ **[Keywords]** What is a Final/Sealed class?	■ **[Keywords]** What is the 'super' keyword?
■ **[Polymorphism]** What is operator overloading?	■ **[Virtual]** What is function overriding with virtual table (vtable)?
■ **[Virtual]** What is a vtable / vptr?	■ **[Virtual]** What is dynamic dispatch?
■ **[Virtual]** What is early binding vs late binding?	■ **[Generics]** What is a template/generic in OOP?
■ **[Exception]** Checked vs Unchecked Exceptions	■ **[Memory]** What is garbage collection?
■ **[Memory]** What is a memory leak?	■ **[Memory]** What is shallow copy vs deep copy?
■ **[Relationships]** What is Association, Aggregation, Composition?	■ **[Relationships]** Difference between Composition and Aggregation
■ **[Design]** What is Cohesion and Coupling?	■ **[Design]** SOLID Principles – explain each
■ **[Design]** Single Responsibility Principle	■ **[Design]** Open/Closed Principle

■ **[Design]** Liskov Substitution Principle	■ **[Design Patterns]** What is a Design Pattern? Name common ones.
■ **[Design Patterns]** Singleton Design Pattern	■ **[Design Patterns]** Factory Design Pattern
■ **[Architecture]** What is MVC pattern?	■ **[OOP Concepts]** What is the difference between IS-A and HAS-A relationship?
■ **[Polymorphism]** Can we override static methods?	■ **[Polymorphism]** Can we override private methods?
■ **[Polymorphism]** Difference between method hiding and method overriding	■ **[Constructors]** Can we call a virtual function from constructor?
■ **[Interface]** Can Interface have a constructor?	■ **[Access Modifiers]** What is friend function in C++?
■ **[Exception]** What is a Custom Exception?	■ **[Design]** Interface Segregation Principle
■ **[Design]** Dependency Inversion Principle	■ **[Design Patterns]** Observer Design Pattern
■ **[Advanced OOP]** What is an inner class / nested class?	■ **[Advanced OOP]** What is an anonymous class?
■ **[Design Patterns]** Decorator Design Pattern	■ **[Design Patterns]** Strategy Design Pattern

ULTIMATE TCS PRIME CHECKLIST

TIER 1	**Absolutely Mandatory**	These questions appear in almost every TCS interview/test. You CANNOT afford to miss these. Master them completely before anything else.	0 items
TIER 2	**Very Important**	High probability questions. Appear frequently in Digital and Prime rounds. Solve all Tier 1 first then tackle these.	0 items
TIER 3	**Good to Know**	Medium frequency. Differentiates average from strong candidates. Cover after Tier 1+2 are solid.	0 items
TIER 4	**Rarely Asked**	Low frequency. Mostly Prime-level. Cover only if time permits.	0 items

COVERAGE PERCENTAGE ANALYSIS

Total questions in this guide: **698** | Tier 1: **151** | Tier 2: **331** | Tier 3: **187** | Tier 4: **29**

Milestone	Questions Covered	% of Total	Primary Focus	Expected Coverage
Top 25	25	3.6%	Tier 1 DSA Core + Top SQL + DBMS Basics	~60-70% of NQT coding round
Top 50	50	7.2%	Tier 1 All Subjects + Tier 2 DSA	~75-80% of NQT + Digital Screening
Top 100	100	14.3%	Tier 1+2 All Subjects	~85-90% of NQT+Digital rounds
Top 200	200	28.7%	Tier 1+2+3 All Subjects	~92-95% of all TCS rounds
COMPLETE	698	100.0%	Full Guide	99%+ coverage of reported questions

Visual Coverage Breakdown by Subject:

DSA	313 (44.8%)
SQL	80 (11.5%)
DBMS	83 (11.9%)
OS	77 (11.0%)
CN	75 (10.7%)
OOP	70 (10.0%)

Final Note: *This guide is compiled from candidate experiences shared on GeeksforGeeks, PrepInsta, FacePrep, FreshersNow, Reddit, GitHub repositories, YouTube interview walkthroughs, and LinkedIn posts between 2021-2026. Questions marked "Very High" frequency have been reported by 20+ candidates independently. Good luck with your preparation!*